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SUMMARY

In this report, the results of open water test, the resistance tests with/without PSV (Pre-Shrouded Vanes) and self-propulsion tests with/without PSV are presented.

The ship model（ was made of wood with a scale ratio of 31.00. Its surface was painted smoothly.

In order to stimulate turbulence, two trip wires of 1.0 mm diameter were fitted to the model at station 19 and bulbous bow.

The ship model was equipped with one rudder in all tests. The design propeller was used in self-propulsion tests. All tests were carried out in the Deep Water Towing Tank of

All the results of the model tests were converted into those of full scale ship in seawater of 15℃.

1 TEST METHOD

1.1 Propeller open water test

The test method is to keep the rate of revolution constant while the speed of advance is varied so that a loading range of the propeller is examined. Thrust, torque and rate of revolution are measured on the shaft behind the propeller model.

Thrust and torque measured during the test are converted into non-dimensional coefficients which are defined as:

$$K_T = \frac{T_m}{\rho_m {n_m}^2 D_{Pm}^4} \tag{1.1.1}$$

$$K_Q = \frac{Q_m}{\rho_m {n_m}^2 D_{Pm}^5} \tag{1.1.2}$$

$$J = \frac{V_{Am}}{n_m D_{Pm}} \tag{1.1.3}$$

$$\eta_o = \frac{J}{2\pi} \cdot \frac{K_T}{K_Q} \tag{1.1.4}$$

where

T_m and Q_m	:	thrust(N) and torque（N m）of model propeller
V_{Am} and n_m	:	speed of advance(m/s) and propeller rate of revolution(rps)
D_{Pm}	:	diameter(m) of model propeller
ρ_m	:	mass density(kg/m^3) of tank water
J	:	advance ratio
K_T and K_Q	:	thrust coefficient and torque coefficient in open water
η_o	:	propeller efficiency in open water

1.2 Resistance test

The results of model tests were converted into those of full scale ship in seawater of 15℃.The friction resistance coefficient was calculated according to the ITTC 1957 model-ship correlation line.

$$C_f = \frac{0.075}{(\lg R_n - 2)^2} \tag{1.2.1}$$

$$R_n = \frac{VL_{WL}}{\nu} \tag{1.2.2}$$

$$C_{tm} = \frac{R_m}{0.5\rho_m V_m^2 S_m} \tag{1.2.3}$$

$$C_{tm} = (1+k)C_{fm} + C_{rm} \tag{1.2.4}$$

If $F_{ns} = F_{nm}$,

$$C_{rm} = C_{rs} \tag{1.2.5}$$

$$C_{ts} = \frac{S_s + S_{BK}}{S_s}\left[(1+k)C_{fs} + \Delta C_F\right] + C_{rs} + C_{AA} \tag{1.2.6}$$

$$R_s = 0.5\rho_s (0.5144V_s)^2 S_s \cdot C_{ts} \tag{1.2.7}$$

$$P_{ES} = 0.5144 V_S R_S \tag{1.2.8}$$

where

V_m and V_s	:	speed of model(m/s) and full scale ship(kn)
R_m and R_s	:	total resistance of model(N) and full scale ship(kN)
S_m and S_s	:	wetted surface of model (m^2) and full scale ship(m^2)
ρ_m and ρ_s	:	mass density(kg/m^3) of tank water and sea water
S_{BK}	:	area of bilge keel
P_{ES}	:	effective power(kW) of full scale ship
C_{tm} and C_{ts}	:	total resistance coefficient of model and full scale ship
C_{fm} and C_{fs}	:	frictional resistance coefficient of model and full scale ship
C_{rm} and C_{rs}	:	residuary resistance coefficient of model and full scale ship
ΔC_F	:	roughness coefficient
C_{AA}	:	air resistance coefficient
k	:	Form factor

1.3 Self-propulsion test

The ship model is equipped with propelling machinery and working propeller. In the self-propulsion(British way), the model was towed at speeds giving the same Froude number as for the full scale ship. The model was towed at 5~6 speed range and the propeller rate of revolutions was varied 4~5 times for each speed, so could get T_m, Q_m, n_m, V_m and F_D.

$$F_D = \frac{1}{2}\rho_m V_m^2 S_m \left\{(1+k)C_{fm} - \frac{S_s + S_{BK}}{S_s}\left[(1+k)C_{fs} + \Delta C_F\right]\right\} \tag{1.3.1}$$

where

F_D	:	towing force(N)

$$K_{Tm} = \frac{T_m}{\rho_m n_m^2 D_{Pm}^4} \tag{1.3.2}$$

$$K_{Qm} = \frac{Q_m}{\rho_m n_m^2 D_{Pm}^5} \tag{1.3.3}$$

where

T_m and Q_m : thrust(N) and torque(N·m) of model propeller behind model ship

n_m : rate of revolution (rps) of model propeller behind model ship

K_{Tm} and K_{Qm} : thrust coefficient and torque coefficient of model propeller behind model ship

With K_{Tm} as input data, the advance ratio and torque coefficient are derived from the open water characteristics of the model propeller.

$$t_m = \frac{T_m + F_D - R_m}{T_m} \tag{1.3.4}$$

$$w_m = 1 - \frac{JD_{Pm}n_m}{V_m} \tag{1.3.5}$$

$$\eta_{hm} = \frac{1 - t_m}{1 - w_m} \tag{1.3.6}$$

$$\eta_{rm} = \frac{K_Q}{K_{Qm}} \tag{1.3.7}$$

$$\eta_{om} = \frac{J}{2\pi} \cdot \frac{K_T}{K_Q} \tag{1.3.8}$$

where

t_m : thrust deduction fraction

w_m : wake fraction

η_{hm} : hull efficiency

η_{rm} : relative rotative efficiency

η_{om} : propeller efficiency in open water

η_{Dm} : Propulsive efficiency

2 PRINCIPAL DIMENSIONS AND COEFFICIENTS

Ship model : Scale ratio : 31.00

DESIGNATION	SYMBOL	Design
Length overall	Loa(m)	224.92
Length on waterline	L_{WL}(m)	220.70
Length between perpendicular	L_{PP}(m)	217.00
Breadth moulded on WL	B(m)	32.25
Wetted surface area	S(m^2)	11044
Displacement volume moulded	(m^3)	73346
Draft moulded on FP	T_F(m)	12.20
Draft moulded on AP	T_A(m)	12.20
Proj. area above WL	At(m^2)	646
Area of bilge keels	S_{bk}(m^2)	110
Block coefficient	CB	0.8591
LCB position of midship	LCB(lpp%)	2.85%
Length-breadth ratio	L_{PP}/B	6.7287
Breadth-draft ratio	B/T	2.6434

DESIGNATION	SYMBOL	Scantling
Length overall	Loa(m)	224.92
Length on waterline	L_{WL}(m)	220.70
Length between perpendicular	L_{PP}(m)	217.00
Breadth moulded on WL	B(m)	32.25
Wetted surface area	S(m^2)	11985
Displacement volume moulded	(m^3)	86941
Draft moulded on FP	T_F(m)	14.25
Draft moulded on AP	T_A(m)	14.25
Proj. area above WL	At(m^2)	589
Area of bilge keels	S_{bk}(m^2)	110
Block coefficient	CB	0.8718
LCB position of midship	LCB(lpp%)	2.28%
Length-breadth ratio	L_{PP}/B	6.7287
Breadth-draft ratio	B/T	2.2632

DESIGNATION	SYMBOL	Ballast
Length overall	Loa(m)	224.92
Length on waterline	L_{WL}(m)	214.86
Length between perpendicular	L_{PP}(m)	217.00
Breadth moulded on WL	B(m)	32.25
Wetted surface area	S(m^2)	8225
Displacement volume moulded	(m^3)	36134
Draft moulded on FP	T_F(m)	5.00
Draft moulded on AP	T_A(m)	7.80
Proj. area above WL	At(m^2)	852
Area of bilge keels	S_{bk}(m^2)	110
Block coefficient	CB	0.8068
LCB position of midship	LCB(lpp%)	1.31%
Length-breadth ratio	L_{PP}/B	6.7287
Breadth-draft ratio	B/T	5.0391

DESIGNATION	SYMBOL	Heavy Ballast
Length overall	Loa(m)	224.92
Length on waterline	L_{WL}(m)	217.91
Length between perpendicular	L_{PP}(m)	217.00
Breadth moulded on WL	B(m)	32.25
Wetted surface area	S(m^2)	8485
Displacement volume moulded	(m^3)	39024
Draft moulded on FP	T_F(m)	4.40
Draft moulded on AP	T_A(m)	9.40
Proj. area above WL	At(m^2)	836
Area of bilge keels	S_{bk}(m^2)	110
Block coefficient	CB	0.8082
LCB position of midship	LCB(lpp%)	-1.15%
Length-breadth ratio	L_{PP}/B	6.7287
Breadth-draft ratio	B/T	4.6739

3 PARTICULARS OF MAIN ENGINE AND PROPELLER

MAIN ENGINE :

Type	Wartsila 5RT-flex58T-D
S.M.C.R ×RPM	9500 kW x 105 r/min
C.S.R ×RPM	8075 kW ×99.5 r/min
Transmission efficiency η_S	0.99

PROPELLER

Scale ratio: 31.00

Particulars	Symbol		
		Model	Full Scale
Diameter	D_P m	0.2194	6.800
Blade area ratio	A_E/A_O	0.45	
Pitch ratio	$(P/D_{PS})_{0.75R}$	0.6742	
Number of blades	Z	4	
Direction of rotation		Right-handed	

4 TEST RESULTS

4.1 Propeller open water test

The design propeller open water characteristics were shown in Table1 and Figure1 in Enclosures.

4.2 Resistance test

The results of resistance test were shown in Table 2,5,8,11,14,17,20,23 in Enclosures.

4.3 Self-propulsion test

The results of self-propulsion test were shown in Table3,6,9,12,15,18,21,24 in Enclosures.

5 SPEED/POWER TRIAL PREDICTION

5.1 Method and Conversion

Without PSV, the full-scale wake was calculate from the wake fraction and thrust deduction fraction according to:

$$w_s = (t_m + 0.04) + (w_m - t_m - 0.04)\frac{(1+k)C_{fS} + \Delta C_F}{(1+k)C_{fm}} \tag{5.1.1}$$

With PSV, the full-scale wake was calculate from the wake fraction and thrust deduction fraction according to:

$$Ei = (1-w_s)/(1-w_{ms}) \quad \text{(Ei method)} \tag{5.1.2}$$

and if $w_m \le w_s$, w_s will be set equal to w_m

$$t_m = t_s \tag{5.1.3}$$

$$\eta_{hs} = \frac{1-t_s}{1-w_s} \tag{5.1.4}$$

$$\eta_{rm} = \eta_{rs} \tag{5.1.5}$$

$$\eta_{DS} = \eta_{hs} \cdot \eta_{os} \cdot \eta_{rs} \tag{5.1.6}$$

$$C_{ts} = \frac{S_s + S_{BK}}{S_s}(C_{fs} + \Delta C_F) + C_{AA} + C_{rs} \tag{5.1.7}$$

$$R_s = 0.5\rho_s(0.5144V_s)^2 S_s \cdot C_{ts} \tag{5.1.8}$$

$$P_{ES} = 0.5144V_s R_s \tag{5.1.9}$$

$$P_{DS} = \frac{P_{ES}}{\eta_{DS}} \tag{5.1.10}$$

$$P_{DT} = P_{DS} \cdot C_P \tag{5.1.11}$$

$$N_T = N_S \cdot C_N \tag{5.1.12}$$

where

t_m and t_s	:	thrust deduction fraction of model and full scale ship
w_m and w_s	:	wake fraction of model and full scale ship
w_{ms}	:	model wake fraction with PSV
η_{hm} and η_{hs}	:	hull efficiency of model and full scale ship
η_{rm} and η_{rs}	:	relative rotative efficiency of model and full scale ship
η_{os}	:	propeller efficiency in open water of full scale ship
η_{DS}	:	propulsive efficiency of full scale ship
P_{DS} and P_{DT}	:	Delivered power at propeller of full scale ship
C_P	:	correction factor for delivered power

C_N : correction factor for revolution

The open water characteristics of design full scale propeller are determined by correcting the scale effects of model according to 1978 ITTC Performance Prediction Method.

$$K_{Ts} = K_{Tm} - \Delta K_T \tag{5.1.13}$$

$$K_{Qs} = K_{Qm} - \Delta K_Q \tag{5.1.14}$$

$$\Delta K_T = -0.3\Delta C_D \left(\frac{P}{D}\right)\left(\frac{c}{D}\right)Z \tag{5.1.15}$$

$$\Delta K_Q = 0.25\Delta C_D \left(\frac{c}{D}\right)Z \tag{5.1.16}$$

$$\Delta C_D = C_{Dm} - C_{Ds} \tag{5.1.17}$$

$$C_{Dm} = 2\left(1+2\frac{t}{c}\right)\left[\frac{0.044}{R_n^{1/6}} - \frac{5}{R_n^{2/3}}\right] \tag{5.1.18}$$

$$C_{Ds} = 2\left(1+2\frac{t}{c}\right)\left[1.89 + 1.62\log\frac{c}{K_p}\right]^{-2.5} \tag{5.1.19}$$

where

K_{Tm} and K_{Ts} : thrust coefficient of model and full scale propeller

K_{Qm} and K_{Qs} : torque coefficient of model and full scale propeller

C_{Dm} and C_{Ds} : resistance coefficient of model and full scale propeller

D : diameter of propeller

Z : number of propeller blades

P : pitch of propeller at radius 0.75D/2

c : cord length of propeller blades at radius 0.75D/2

t : maximum blade thickness at radius 0.75D/2

R_n : the local Reynolds number at radius 0.75D/2

K_p : full scale blade roughness, which is set to $K_p = 30\times10^{-6}$ m

5.2 Speed/power trial prediction

Speed prediction made under the condition of calm sea, sufficient water depth and clean bottom were shown in Table 4,7,10,13,16,19,22,25 and Fig2,3,4,5,6,7,8,9.

Ship speed prediction under given power

Load condition	With/ Without PSV	Delivered power at propeller P_{DT}(kW)	Ship speed V_S（kn）	Rotational speed N_T（r/min）
Design draft T=12.20/12.20m	without PSV	6952（15% S.M.）	14.25	99.3
	with PSV	6952（15% S.M.）	14.52	97.7
Scantling draft T=14.25/14.25m	without PSV	6952（15% S.M.）	13.71	98.9
	with PSV	6952（15% S.M.）	13.85	97.2
Ballast draft T=5.00/7.80m	without PSV	6952（15% S.M.）	14.83	100.1
	with PSV	6952（15% S.M.）	15.02	97.5
Heavy ballast draft T=4.40/9.40m	without PSV	6952（15% S.M.）	14.67	99.5
	with PSV	6952（15% S.M.）	14.89	98.7

Load condition	With/ Without PSV	Delivered power at propeller P_{DT}(kW)	Ship speed V_S（kn）	Rotational speed N_T（r/min）
Design draft T=12.20/12.20m	without PSV	7994（0% S.M.）	14.79	104.0
	with PSV	7994（0% S.M.）	15.02	102.2
Scantling draft T=14.25/14.25m	without PSV	7994（0% S.M.）	14.26	103.5
	with PSV	7994（0% S.M.）	14.39	101.8
Ballast draft T=5.00/7.80m	without PSV	7994（0% S.M.）	15.36	104.4
	with PSV	7994（0% S.M.）	15.54	102.6
Heavy ballast draft T=4.40/9.40m	without PSV	7994（0% S.M.）	15.23	104.1
	with PSV	7994（0% S.M.）	15.43	103.3

Load Condition	V_S （kn）	Without PSV		with PSV		Energy saving amount $1-P_{DT2}/P_{DT1}$
		Nt (rpm)	Pdt （kW）	Nt (rpm)	Pdt （kW）	
Design draft	11.50	79.4	3504	75.8	3295	5.9%
	12.00	82.8	3968	79.4	3772	4.9%
	12.50	86.3	4492	82.9	4283	4.7%
	13.00	89.9	5083	86.4	4815	5.3%
	13.50	93.6	5755	89.8	5397	6.2%
	14.00	97.4	6526	93.4	6078	6.9%
	14.50	101.5	7417	97.5	6909	6.9%
	15.00	106.0	8461	102.0	7941	6.1%
	15.50	110.9	9724	106.8	9231	5.1%

Load Condition	V_S （kn）	Without PSV		with PSV		Energy saving amount $1-P_{DT2}/P_{DT1}$
		Nt (rpm)	Pdt （kW）	Nt (rpm)	Pdt （kW）	
Scantling draft	11.00	78.6	3472	76.2	3327	4.2%
	11.50	82.3	4012	80.1	3836	4.4%
	12.00	85.9	4591	83.9	4379	4.6%
	12.50	89.6	5199	87.4	4968	4.4%
	13.00	93.3	5855	90.8	5621	4.0%
	13.50	97.2	6599	94.4	6363	3.6%
	14.00	101.3	7470	98.4	7226	3.3%
	14.50	105.6	8507	102.7	8223	3.3%
	15.00	110.1	9729	107.2	9359	3.8%

Load Condition	V_S （kn）	Without PSV		with PSV		Energy saving amount $1-P_{DT2}/P_{DT1}$
		Nt (rpm)	Pdt （kW）	Nt (rpm)	Pdt （kW）	
Ballast draft	12.00	77.2	3285	73.7	3059	6.9%
	12.50	81.0	3784	76.8	3525	6.8%
	13.00	84.8	4335	80.1	4041	6.8%
	13.50	88.9	4934	83.9	4643	5.9%
	14.00	93.1	5599	88.1	5324	4.9%
	14.50	97.3	6365	92.6	6067	4.7%
	15.00	101.5	7260	97.3	6907	4.9%
	15.50	105.5	8299	102.2	7902	4.8%
	16.00	109.4	9488	107.2	9074	4.4%

Load Condition	V_S （kn）	Without PSV		with PSV		Energy saving amount $1-P_{DT2}/P_{DT1}$
		Nt (rpm)	Pdt （kW）	Nt (rpm)	Pdt （kW）	
Heavy ballast draft	12.00	78.7	3404	76.6	3249	4.6%
	12.50	82.6	3935	80.2	3723	5.4%
	13.00	86.5	4518	83.8	4242	6.1%
	13.50	90.3	5160	87.6	4837	6.3%
	14.00	94.2	5872	91.5	5515	6.1%
	14.50	98.1	6663	95.5	6284	5.7%
	15.00	102.2	7547	99.6	7158	5.2%
	15.50	106.4	8545	104.0	8150	4.6%
	16.00	110.7	9677	108.4	9294	4.0%

Note:

The speed prediction was made under the condition of calm sea, sufficient water depth and clean bottom.

6 ENCLOSURES

APPENDIX A

OPEN WATER TEST RESULTS

Propeller model No	:	Propeller diameter	: D_{Pm}=0.2194 m
Number of revolution	: n_m=18 rps	Pitch ratio	: $(P/D_{PS})_{0.7R}$=0.6742
Temperature of water	: t_m=19℃	Blade area ratio	: A_E/A_O =0.45
Reynolds number	: R_{nm}=4.754×10^5	Number of blades	: Z=4
		Direction of rotation	: Right

Table 1 Propeller's open-water performance

Model Test			
J	K_T	$10K_Q$	η_o
0.1000	0.2626	0.2566	0.1629
0.1500	0.2459	0.2434	0.2412
0.2000	0.2286	0.2299	0.3165
0.2500	0.2108	0.2161	0.3882
0.3000	0.1925	0.2017	0.4556
0.3500	0.1736	0.1868	0.5176
0.4000	0.1542	0.1713	0.5730
0.4500	0.1342	0.1550	0.6203
0.5000	0.1138	0.1378	0.6567
0.5500	0.0928	0.1198	0.6781
0.6000	0.0713	0.1008	0.6762
0.6500	0.0494	0.0806	0.6340
0.7000	0.0270	0.0593	0.5070
Full scale			
J	K_{TS}	$10K_{QS}$	η_{oS}
0.1000	0.2629	0.2533	0.1651
0.1500	0.2462	0.2401	0.2447
0.2000	0.2289	0.2267	0.3215
0.2500	0.2111	0.2128	0.3947
0.3000	0.1927	0.1984	0.4637
0.3500	0.1738	0.1835	0.5276
0.4000	0.1544	0.1680	0.5852
0.4500	0.1345	0.1517	0.6349
0.5000	0.1140	0.1346	0.6743
0.5500	0.0931	0.1165	0.6991
0.6000	0.0716	0.0975	0.7015
0.6500	0.0497	0.0774	0.6643
0.7000	0.0273	0.0560	0.5418

Note:

$$R_{nm} = \frac{C_{m0.75}}{\nu_m}\sqrt{V_{Am}^2 + (0.75\pi n_m D_{Pm})^2} \quad (V_{Am}=0)$$

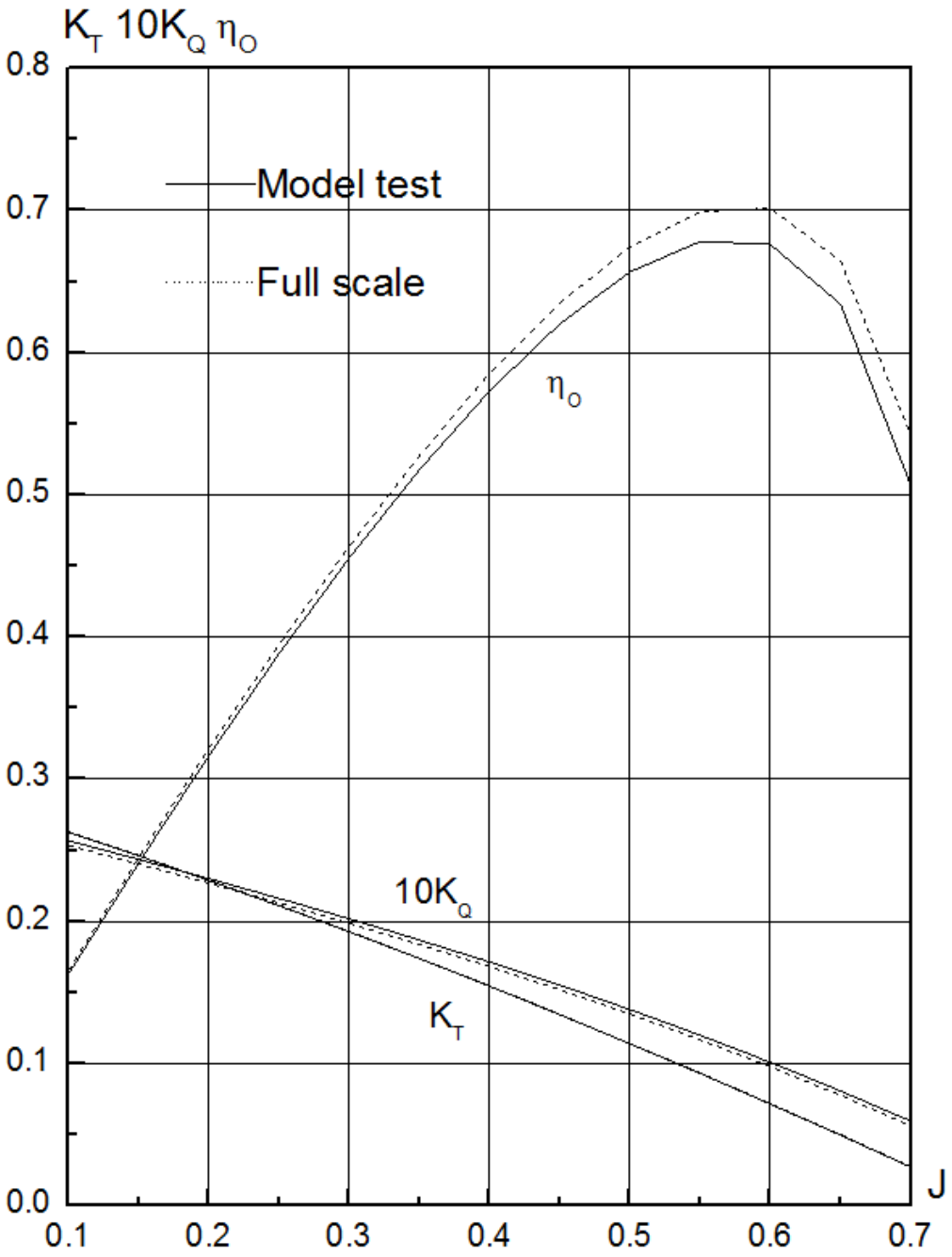


Fig.1 OPEN WATER PERFORMANCE CHARACTERISTICS

RESISTANCE TEST RESULTS TEST OF DESIGN DRAFT WITHOUT PSV

Ship model No	:		Length of waterline in general	:	L_{WL} =220.70 m
Condition	:	Design draft	Draught (F.P)	:	T_F=12.20 m
Temperature of water	:	t_m=11.0℃	Draught (A.P)	:	T_A=12.20 m
Date of tests	:	2017-3-20	Displacement	:	▽ =73346 m^3
Appendages	:	Rudder	Wetted surface	:	S =11044 m^2
			Area of bilge keel	:	S_{BK} =110 m^2
			Proj. area above water line	:	A_T =646 m^2

Form factor : 1+k=1

Roughness coefficient : $\Delta C_F = -0.15 \times 10^{-3}$

Air resistance coefficient : $C_{AA} = 0.058 \times 10^{-3}$

Table 2

V_S（kn）	F_n	C_{tm}（10^{-3}）	C_{fm}（10^{-3}）	C_{rm}（10^{-3}）
11.50	0.1272	3.9730	3.2893	0.6837
12.00	0.1327	3.9517	3.2640	0.6877
12.50	0.1382	3.9388	3.2400	0.6988
13.00	0.1437	3.9320	3.2172	0.7148
13.50	0.1493	3.9305	3.1955	0.7350
14.00	0.1548	3.9358	3.1747	0.7611
14.50	0.1603	3.9597	3.1549	0.8048
15.00	0.1659	4.0145	3.1359	0.8786
15.50	0.1714	4.1067	3.1178	0.9889

V_S（kn）	C_{rs}（10^{-3}）	C_{ts}（10^{-3}）	R_S（kN）	P_{ES}（kW）
11.50	0.6837	2.1190	420.26	2486
12.00	0.6877	2.1150	456.74	2819
12.50	0.6988	2.1185	496.42	3192
13.00	0.7148	2.1273	539.15	3605
13.50	0.7350	2.1406	585.05	4063
14.00	0.7611	2.1600	634.89	4572
14.50	0.8048	2.1973	692.83	5168
15.00	0.8786	2.2650	764.27	5897
15.50	0.9889	2.3695	853.72	6807

SELF-PROPULSION TEST RESULTS OF DESIGN DRAFT WITHOUT PSV

Ship model No	:		Length of waterline in general	:	L_{WL} =220.70 m
Propeller model No	:		Draught (F.P)	:	T_F=12.20 m
Condition	:	Design draft	Draught (A.P)	:	T_A=12.20 m
Temperature of water	:	t_m=11.0℃	Displacement	:	▽ =73346 m^3
Date of tests	:	2017-3-20	Wetted surface	:	S =11044 m^2
Appendages	:	Rudder	Area of bilge keel	:	S_{BK} =110 m^2
			Proj. area above water line	:	A_T =646 m^2
Form factor	:	1+k=1	Propeller diameter	:	D_{PS} =6.800m
Roughness coefficient	:	$\Delta C_F = -0.15 \times 10^{-3}$	Pitch ratio	:	$(P/D_{PS})_{0.75R}$=0.6742
Air resistance coefficient	:	$C_{AA} = 0.058 \times 10^{-3}$	Blade area ratio	:	A_E/A_O =0.45
			Number of blades	:	Z=4
			Direction of rotation	:	Right

Table 3

V_m (m/s)	R_m (N)	F_D (N)	T_m (N)	Q_m (N.m)	N_m (rps)
1.063	25.770	12.494	17.144	0.442	7.015
1.155	30.185	14.490	20.597	0.523	7.671
1.247	35.133	16.618	24.626	0.620	8.353
1.340	40.833	18.904	29.611	0.738	9.090
1.432	48.391	21.291	35.910	0.886	9.908
V_m (m/s)	$t_m(t_s)$	w_m	$\eta_{rm}(\eta_{rs})$	η_{om}	J_t
1.063	0.226	0.407	0.950	0.583	0.4099
1.155	0.238	0.406	0.964	0.581	0.4081
1.247	0.248	0.405	0.971	0.578	0.4048
1.340	0.259	0.406	0.975	0.572	0.3989
1.432	0.245	0.407	0.980	0.563	0.3907
V_S(kn)	w_s	η_{hs}	η_{os}	η_{DS}	J_S
11.50	0.324	1.146	0.626	0.681	0.4401
12.50	0.331	1.139	0.621	0.682	0.4352
13.50	0.337	1.134	0.616	0.678	0.4296
14.50	0.344	1.129	0.607	0.669	0.4212
15.50	0.336	1.137	0.603	0.672	0.4168

SPEED/POWER TRIAL PREDICTION OF DESIGN DRAFT WITHOUT PSV

Ship model No	:		Diameter of design propeller	:	D_{PS} =6.800m
Condition	:	Design draft	Pitch ratio	:	$(P/D_{PS})_{0.75R}$=0.6742
Draught (F.P)	:	T_F=12.20 m	Blade area ratio	:	A_E/A_O =0.45
Draught (A.P)	:	T_A=12.20 m	Number of blades	:	Z=4
Displacement	:	∇ =73346 m^3	Direction of rotation	:	Right
Wetted surface	:	S =11044 m^2	Number of propellers	:	1
Area of bilge keel	:	S_{BK} =110 m^2			
Proj. area above water line	:	A_T =646 m^2			
Form factor	:	1+k=1			
Roughness coefficient	:	$\Delta C_F = -0.15 \times 10^{-3}$	Appendage	:	Rudder
Air resistance coefficient	:	$C_{AA} = 0.058 \times 10^{-3}$			
Power correction factor	:	C_P = 0.96			
Revolution correction factor	:	C_N = 0.99			

Table 4

V_S(kn)	t_s	w_s	η_{hs}	η_{rs}	η_{OS}	η_{DS}	P_{ES}(kW)	P_{DT}(kW)	N_T(rpm)
11.50	0.226	0.324	1.146	0.950	0.626	0.681	2486	3504	79.4
12.00	0.232	0.328	1.142	0.958	0.623	0.682	2819	3968	82.8
12.50	0.238	0.331	1.139	0.964	0.621	0.682	3192	4492	86.3
13.00	0.243	0.334	1.137	0.968	0.619	0.681	3605	5083	89.9
13.50	0.248	0.337	1.134	0.971	0.616	0.678	4063	5755	93.6
14.00	0.256	0.342	1.131	0.973	0.611	0.673	4572	6526	97.4
14.50	0.259	0.344	1.129	0.975	0.607	0.669	5168	7417	101.5
15.00	0.255	0.342	1.132	0.978	0.605	0.669	5897	8461	106.0
15.50	0.245	0.336	1.137	0.980	0.603	0.672	6807	9724	110.9

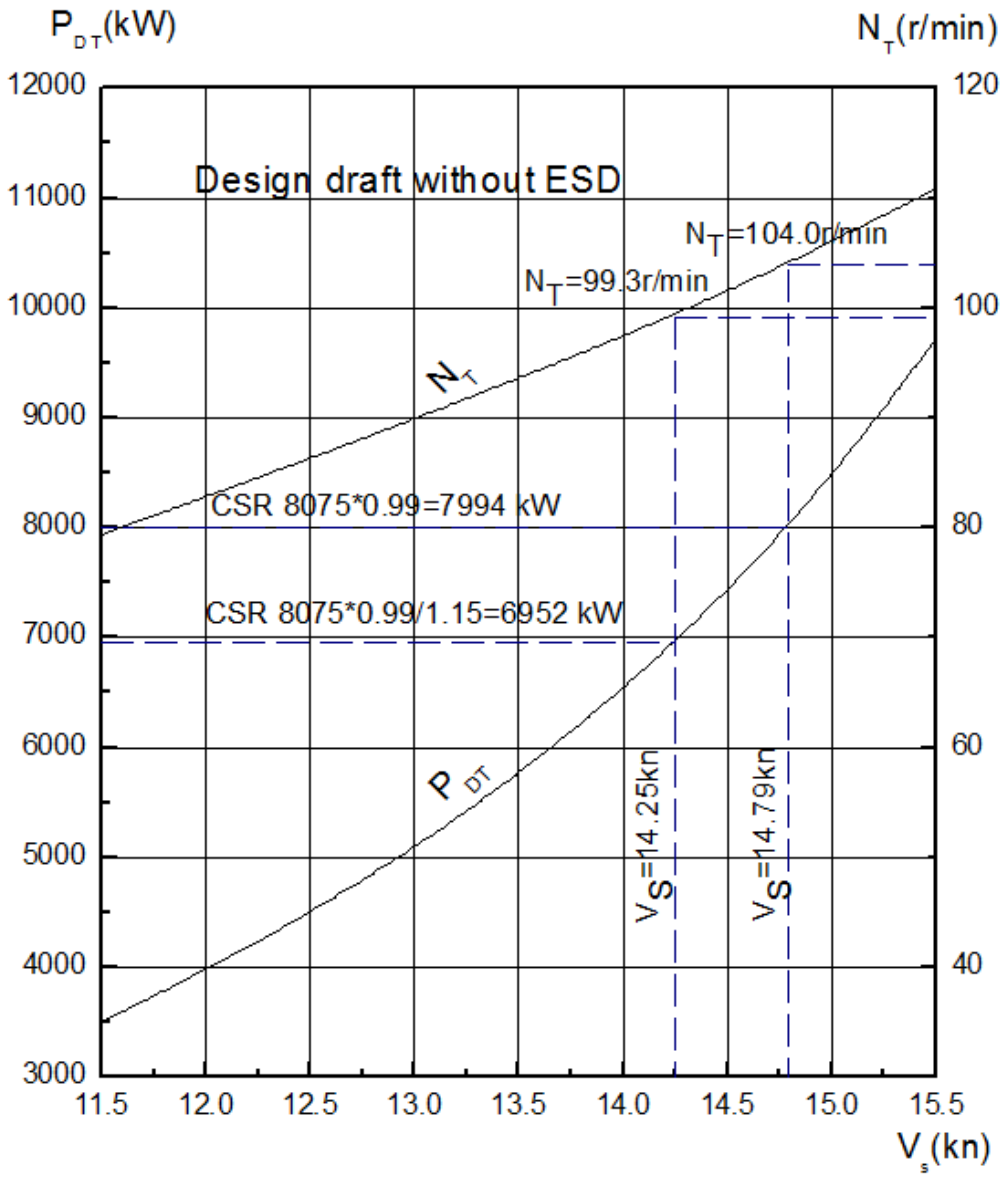


Fig. 2 SPEED/POWER TRIAL PREDICTION CURVES OF DESIGN DRAFT WITHOUT PSV

RESISTANCE TEST RESULTS TEST OF DESIGN DRAFT WITH PSV

Ship model No	:		Length of waterline in general	:	L_{WL} =220.70 m
Condition	:	Design draft	Draught (F.P)	:	T_F=12.20 m
Temperature of water	:	t_m=11.0℃	Draught (A.P)	:	T_A=12.20 m
Date of tests	:	2017-3-20	Displacement	:	▽ =73346 m^3
Appendages	:	Rudder	Wetted surface	:	S =11101 m^2
		PSV	Area of bilge keel	:	S_{BK} =110 m^2
			Proj. area above water line	:	A_T =646 m^2

Form factor	:	1+k=1
Roughness coefficient	:	$\Delta C_F = -0.15 \times 10^{-3}$
Air resistance coefficient	:	$C_{AA} = 0.058 \times 10^{-3}$

Table 5

V_S（kn）	F_n	C_{tm}（10^{-3}）	C_{fm}（10^{-3}）	C_{rm}（10^{-3}）
11.50	0.1272	3.9812	3.2893	0.6919
12.00	0.1327	3.9620	3.2640	0.6980
12.50	0.1382	3.9451	3.2400	0.7051
13.00	0.1437	3.9296	3.2172	0.7124
13.50	0.1493	3.9192	3.1955	0.7237
14.00	0.1548	3.9199	3.1747	0.7452
14.50	0.1603	3.9443	3.1549	0.7894
15.00	0.1659	4.0037	3.1359	0.8678
15.50	0.1714	4.1013	3.1178	0.9835

V_S（kn）	C_{rs}（10^{-3}）	C_{ts}（10^{-3}）	R_S（kN）	P_{ES}（kW）
11.50	0.6919	2.1268	423.99	2508
12.00	0.6980	2.1249	461.26	2847
12.50	0.7051	2.1244	500.38	3217
13.00	0.7124	2.1245	541.22	3619
13.50	0.7237	2.1289	584.86	4062
14.00	0.7452	2.1437	633.36	4561
14.50	0.7894	2.1816	691.41	5157
15.00	0.8678	2.2538	764.42	5898
15.50	0.9835	2.3637	856.04	6825

SELF-PROPULSION TEST RESULTS OF DESIGN DRAFT WITH PSV

Ship model No	:		Length of waterline in general	:	L_{WL} =220.70 m
Propeller model No	:		Draught (F.P)	:	T_F=12.20 m
Condition	:	Design draft	Draught (A.P)	:	T_A=12.20 m
Temperature of water	:	t_m=11.0℃	Displacement	:	▽ =73346 m^3
Date of tests	:	2017-3-20	Wetted surface	:	S =11101 m^2
Appendages	:	Rudder	Area of bilge keel	:	S_{BK} =110 m^2
			Proj. area above water line	:	A_T =646 m^2
Form factor	:	1+k=1	Propeller diameter	:	D_{PS} =6.800m
Roughness coefficient	:	$\Delta C_F = -0.15 \times 10^{-3}$	Pitch ratio	:	$(P/D_{PS})_{0.75R}$=0.6742
Air resistance coefficient	:	$C_{AA} = 0.058 \times 10^{-3}$	Blade area ratio	:	A_E/A_O =0.45
		Ei=1.12	Number of blades	:	Z=4
			Direction of rotation	:	Right

Table 6

V_m (m/s)	R_m (N)	F_D (N)	T_m (N)	Q_m (N.m)	N_m (rps)
1.063	25.957	12.558	17.241	0.4378	6.769
1.155	30.390	14.565	20.799	0.5204	7.409
1.247	35.214	16.704	24.531	0.6058	8.027
1.340	40.884	19.002	29.281	0.7148	8.720
1.432	48.577	21.401	36.219	0.8740	9.583
V_m (m/s)	$t_m(t_s)$	w_m	$\eta_{rm}(\eta_{rs})$	η_{om}	J_t
1.063	0.223	0.471	0.945	0.551	0.3791
1.155	0.239	0.471	0.957	0.547	0.3762
1.247	0.245	0.472	0.969	0.545	0.3741
1.340	0.253	0.473	0.977	0.540	0.3693
1.432	0.250	0.473	0.982	0.528	0.3588
V_S(kn)	w_s	η_{hs}	η_{os}	η_{DS}	J_S
11.50	0.407	1.310	0.590	0.731	0.4044
12.50	0.407	1.283	0.587	0.721	0.4016
13.50	0.408	1.275	0.585	0.723	0.3997
14.50	0.410	1.266	0.579	0.717	0.3947
15.50	0.410	1.272	0.568	0.710	0.3846

SPEED/POWER TRIAL PREDICTION OF DESIGN DRAFT WITH PSV

Ship model No	:		Diameter of design propeller	:	D_{PS} =6.800m
Condition	:	Design draft	Pitch ratio	:	$(P/D_{PS})_{0.75R}$=0.6742
Draught (F.P)	:	T_F=12.20 m	Blade area ratio	:	A_E/A_O =0.45
Draught (A.P)	:	T_A=12.20 m	Number of blades	:	Z=4
Displacement	:	∇ =73346 m^3	Direction of rotation	:	Right
Wetted surface	:	S =11101 m^2	Number of propellers	:	1
Area of bilge keel	:	S_{BK} =110 m^2			
Proj. area above water line	:	A_T =646 m^2			

Form factor	:	1+k=1			
Roughness coefficient	:	$\Delta C_F = -0.15 \times 10^{-3}$	Appendage	:	Rudder
Air resistance coefficient	:	$C_{AA} = 0.058 \times 10^{-3}$			PSV
		Ei=1.12			

Power correction factor	:	C_P = 0.96
Revolution correction factor	:	C_N = 0.99

Table 7

V_S(kn)	t_s	w_s	η_{hs}	η_{rs}	η_{OS}	η_{DS}	P_{ES}(kW)	P_{DT}(kW)	N_T(rpm)
11.50	0.223	0.407	1.310	0.945	0.590	0.731	2508	3295	75.8
12.00	0.232	0.407	1.295	0.951	0.588	0.725	2847	3772	79.4
12.50	0.239	0.407	1.283	0.957	0.587	0.721	3217	4283	82.9
13.00	0.243	0.408	1.278	0.963	0.586	0.722	3619	4815	86.4
13.50	0.245	0.408	1.275	0.969	0.585	0.723	4062	5397	89.8
14.00	0.250	0.409	1.270	0.973	0.583	0.720	4561	6078	93.4
14.50	0.253	0.410	1.266	0.977	0.579	0.717	5157	6909	97.5
15.00	0.252	0.410	1.267	0.980	0.574	0.713	5898	7941	102.0
15.50	0.250	0.410	1.272	0.982	0.568	0.710	6825	9231	106.8

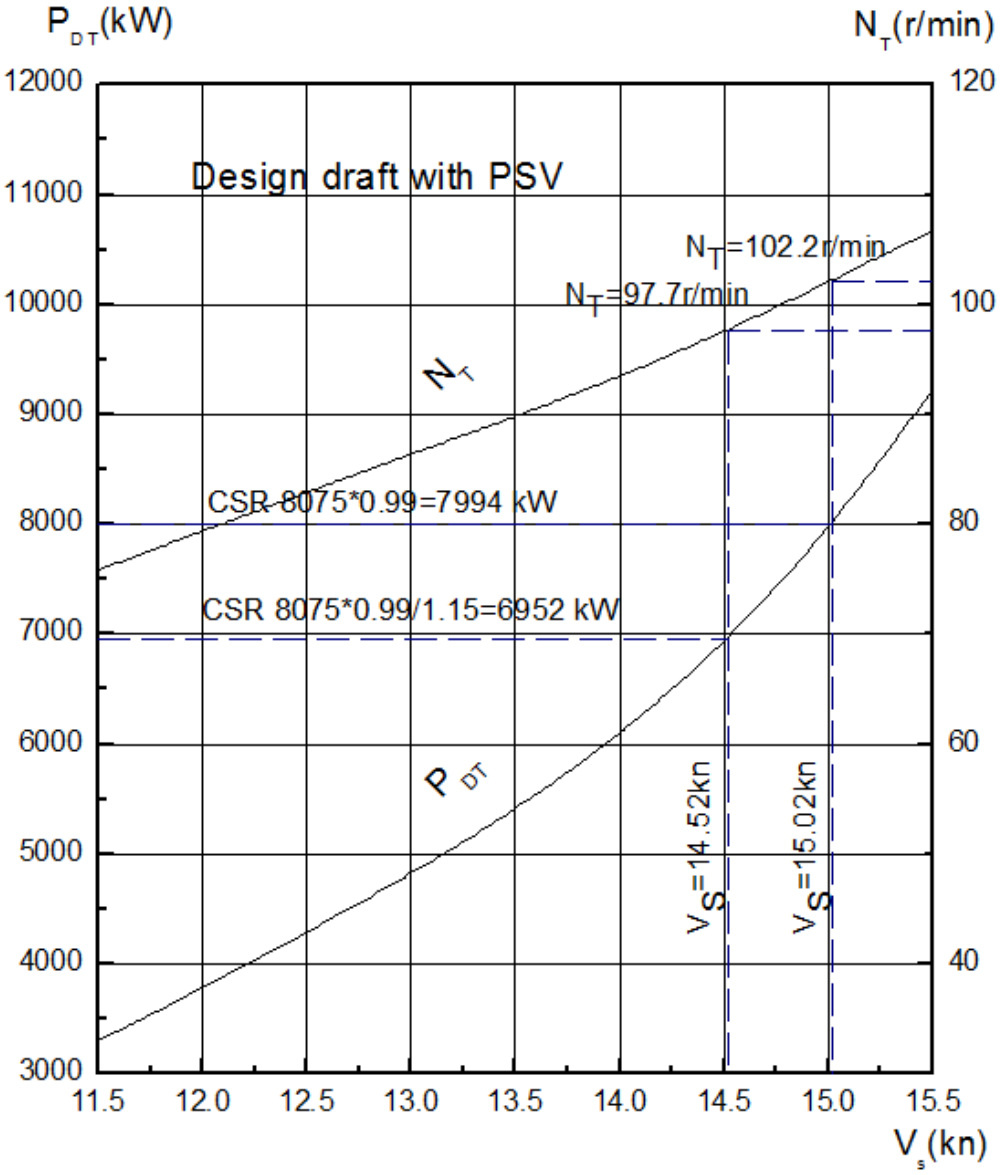


Fig. 3 SPEED/POWER TRIAL PREDICTION CURVES OF DESIGN DRAFT WITH PSV

RESISTANCE TEST RESULTS TEST OF SCANTLING DRAFT WITHOUT PSV

Ship model No	:		Length of waterline in general	:	L_{WL} =220.70 m
Condition	:	Scantling draft	Draught (F.P)	:	T_F=14.25 m
Temperature of water	:	t_m=18.5℃	Draught (A.P)	:	T_A=14.25 m
Date of tests	:	2017-5-17	Displacement	:	▽ =86941 m^3
Appendages	:	Rudder	Wetted surface	:	S =11985 m^2
			Area of bilge keel	:	S_{BK} =110 m^2
			Proj. area above water line	:	A_T =589 m^2

Form factor : 1+k=1

Roughness coefficient : $\Delta C_F = -0.15 \times 10^{-3}$

Air resistance coefficient : $C_{AA} = 0.049 \times 10^{-3}$

Table 8

V_S（kn）	F_n	C_{tm}（10^{-3}）	C_{fm}（10^{-3}）	C_{rm}（10^{-3}）
11.00	0.1216	3.9840	3.1992	0.7848
11.50	0.1272	3.9636	3.1738	0.7898
12.00	0.1327	3.9442	3.1498	0.7944
12.50	0.1382	3.9288	3.1271	0.8017
13.00	0.1437	3.9209	3.1054	0.8155
13.50	0.1493	3.9242	3.0848	0.8394
14.00	0.1548	3.9407	3.0652	0.8755
14.50	0.1603	3.9714	3.0464	0.9250
15.00	0.1659	4.0157	3.0284	0.9873

V_S（kn）	C_{rs}（10^{-3}）	C_{ts}（10^{-3}）	R_S（kN）	P_{ES}（kW）
11.00	0.7848	2.2181	436.80	2472
11.50	0.7898	2.2147	476.66	2820
12.00	0.7944	2.2113	518.22	3199
12.50	0.8017	2.2110	562.25	3615
13.00	0.8155	2.2175	609.91	4079
13.50	0.8394	2.2345	662.76	4602
14.00	0.8755	2.2641	722.19	5201
14.50	0.9250	2.3072	789.46	5888
15.00	0.9873	2.3634	865.42	6678

SELF-PROPULSION TEST RESULTS OF SCANTLING DRAFT WITHOUT PSV

Ship model No	:		Length of waterline in general	:	L_{WL} =220.70 m
Propeller model No	:		Draught (F.P)	:	T_F=14.25 m
Condition	:	Scantling draft	Draught (A.P)	:	T_A=14.25 m
Temperature of water	:	t_m=18.5℃	Displacement	:	▽ =86941 m^3
Date of tests	:	2017-5-17	Wetted surface	:	S =11985 m^2
Appendages	:	Rudder	Area of bilge keel	:	S_{BK} =110 m^2
			Proj. area above water line	:	A_T =589 m^2
Form factor	:	1+k=1	Propeller diameter	:	D_{PS} =6.800m
Roughness coefficient	:	$\Delta C_F = -0.15 \times 10^{-3}$	Pitch ratio	:	$(P/D_{PS})_{0.75R}$=0.6742
Air resistance coefficient	:	$C_{AA} = 0.049 \times 10^{-3}$	Blade area ratio	:	A_E/A_O =0.45
			Number of blades	:	Z=4
			Direction of rotation	:	Right

Table 9

V_m (m/s)	R_m (N)	F_D (N)	T_m (N)	Q_m (N.m)	N_m (rps)
1.016	25.630	11.740	17.739	0.4440	6.948
1.109	30.197	13.734	21.713	0.5393	7.657
1.201	35.231	15.841	25.667	0.6346	8.342
1.294	41.066	18.109	30.358	0.7467	9.065
1.386	48.040	20.483	36.802	0.8968	9.890
V_m (m/s)	$t_m(t_s)$	w_m	$\eta_{rm}(\eta_{rs})$	η_{om}	J_t
1.016	0.217	0.417	0.965	0.561	0.3884
1.109	0.242	0.417	0.970	0.557	0.3852
1.201	0.245	0.411	0.976	0.559	0.3869
1.294	0.244	0.407	0.980	0.558	0.3862
1.386	0.251	0.407	0.985	0.550	0.3786
V_S(kn)	w_s	η_{hs}	η_{os}	η_{DS}	J_S
11.00	0.326	1.161	0.610	0.683	0.4239
12.00	0.340	1.148	0.600	0.669	0.4144
13.00	0.339	1.143	0.600	0.669	0.4138
14.00	0.337	1.140	0.598	0.668	0.4120
15.00	0.342	1.137	0.588	0.659	0.4030

SPEED/POWER TRIAL PREDICTION OF SCANTLING DRAFT WITHOUT PSV

Ship model No	:		Diameter of design propeller	:	D_{PS} =6.800m
Condition	:	Scantling draft	Pitch ratio	:	$(P/D_{PS})_{0.75R}$=0.6742
Draught (F.P)	:	T_F=14.25 m	Blade area ratio	:	A_E/A_O =0.45
Draught (A.P)	:	T_A=14.25 m	Number of blades	:	Z=4
Displacement	:	∇ =86941 m^3	Direction of rotation	:	Right
Wetted surface	:	S =11985 m^2	Number of propellers	:	1
Area of bilge keel	:	S_{BK} =110 m^2			
Proj. area above water line	:	A_T =589 m^2			

Form factor	:	1+k=1			
Roughness coefficient	:	$\Delta C_F = -0.15 \times 10^{-3}$	Appendage	:	Rudder
Air resistance coefficient	:	$C_{AA} = 0.049 \times 10^{-3}$			

Power correction factor	:	C_P = 0.96
Revolution correction factor	:	C_N = 0.99

Table 10

V_S(kn)	t_s	w_s	η_{hs}	η_{rs}	η_{OS}	η_{DS}	P_{ES}(kW)	P_{DT}(kW)	N_T(rpm)
11.00	0.217	0.326	1.161	0.965	0.610	0.683	2472	3472	78.6
11.50	0.231	0.334	1.154	0.967	0.604	0.675	2820	4012	82.3
12.00	0.242	0.340	1.148	0.970	0.600	0.669	3199	4591	85.9
12.50	0.245	0.341	1.145	0.973	0.599	0.668	3615	5199	89.6
13.00	0.245	0.339	1.143	0.976	0.600	0.669	4079	5855	93.3
13.50	0.243	0.337	1.142	0.978	0.600	0.670	4602	6599	97.2
14.00	0.244	0.337	1.140	0.980	0.598	0.668	5201	7470	101.3
14.50	0.247	0.339	1.139	0.982	0.594	0.664	5888	8507	105.6
15.00	0.251	0.342	1.137	0.985	0.588	0.659	6678	9729	110.1

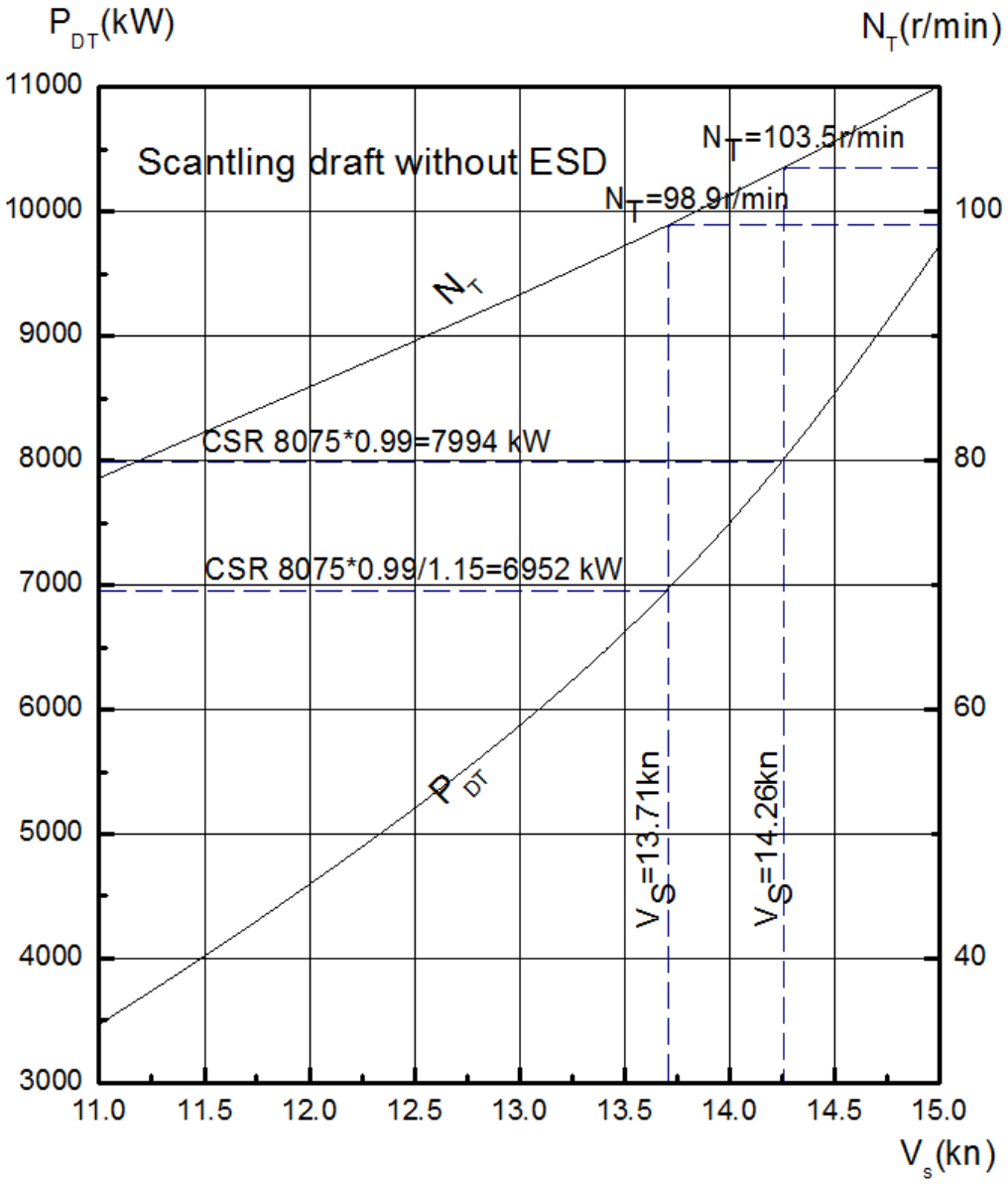


Fig. 4 SPEED/POWER TRIAL PREDICTION CURVES OF SCANTLING DRAFT WITHOUT PSV

RESISTANCE TEST RESULTS TEST OF SCANTLING DRAFT WITH PSV

Ship model No	:		Length of waterline in general	:	L_{WL} =220.70 m
Condition	:	Scantling draft	Draught (F.P)	:	T_F=14.25 m
Temperature of water	:	t_m=18.5℃	Draught (A.P)	:	T_A=14.25 m
Date of tests	:	2017-5-17	Displacement	:	▽ =86941 m^3
Appendages	:	Rudder	Wetted surface	:	S =12042 m^2
		PSV	Area of bilge keel	:	S_{BK} =110 m^2
			Proj. area above water line	:	A_T =589 m^2

Form factor	:	1+k=1
Roughness coefficient	:	$\Delta C_F = -0.15 \times 10^{-3}$
Air resistance coefficient	:	$C_{AA} = 0.049 \times 10^{-3}$

Table 11

V_S（kn）	F_n	C_{tm}（10^{-3}）	C_{fm}（10^{-3}）	C_{rm}（10^{-3}）
11.00	0.1216	3.9161	3.1992	0.7169
11.50	0.1272	3.9111	3.1738	0.7373
12.00	0.1327	3.8981	3.1498	0.7483
12.50	0.1382	3.8858	3.1271	0.7587
13.00	0.1437	3.8829	3.1054	0.7775
13.50	0.1493	3.8983	3.0848	0.8135
14.00	0.1548	3.9344	3.0652	0.8692
14.50	0.1603	3.9860	3.0464	0.9396
15.00	0.1659	4.0485	3.0284	1.0201

V_S（kn）	C_{rs}（10^{-3}）	C_{ts}（10^{-3}）	R_S（kN）	P_{ES}（kW）
11.00	0.7169	2.1499	425.38	2407
11.50	0.7373	2.1619	467.51	2766
12.00	0.7483	2.1649	509.76	3147
12.50	0.7587	2.1678	553.86	3561
13.00	0.7775	2.1792	602.23	4027
13.50	0.8135	2.2083	658.11	4570
14.00	0.8692	2.2575	723.51	5210
14.50	0.9396	2.3215	798.13	5953
15.00	1.0201	2.3959	881.50	6802

SELF-PROPULSION TEST RESULTS OF SCANTLING DRAFT WITH PSV

Ship model No	:		Length of waterline in general	:	L_{WL} =220.70 m
Propeller model No	:		Draught (F.P)	:	T_F=14.25 m
Condition	:	Scantling draft	Draught (A.P)	:	T_A=14.25 m
Temperature of water	:	t_m=18.5℃	Displacement	:	▽ =86941 m^3
Date of tests	:	2017-5-17	Wetted surface	:	S =12042 m^2
Appendages	:	Rudder	Area of bilge keel	:	S_{BK} =110 m^2
		PSV	Proj. area above water line	:	A_T =589 m^2
Form factor	:	1+k=1	Propeller diameter	:	D_{PS} =6.800m
Roughness coefficient	:	ΔC_F =-0.15×10^{-3}	Pitch ratio	:	$(P/D_{PS})_{0.75R}$=0.6742
Air resistance coefficient	:	C_{AA} =0.049×10^{-3}	Blade area ratio	:	A_E/A_O =0.45
		Ei=1.12	Number of blades	:	Z=4
			Direction of rotation	:	Right

Table 12

V_m (m/s)	R_m (N)	F_D (N)	T_m (N)	Q_m (N.m)	N_m (rps)
1.016	25.313	11.796	17.885	0.442	6.823
1.109	29.987	13.799	21.423	0.528	7.511
1.201	35.056	15.916	25.607	0.627	8.137
1.294	41.196	18.195	30.586	0.744	8.833
1.386	48.662	20.581	37.076	0.886	9.636
V_m (m/s)	$t_m(t_s)$	w_m	$\eta_{rm}(\eta_{rs})$	η_{om}	J_t
1.016	0.244	0.455	0.965	0.540	0.3698
1.109	0.244	0.443	0.972	0.546	0.3748
1.201	0.253	0.455	0.973	0.537	0.3669
1.294	0.248	0.459	0.976	0.530	0.3610
1.386	0.243	0.462	0.988	0.521	0.3529
V_S(kn)	w_s	η_{hs}	η_{os}	η_{DS}	J_S
11.00	0.390	1.239	0.581	0.694	0.3960
12.00	0.376	1.212	0.586	0.690	0.4008
13.00	0.389	1.224	0.578	0.688	0.3931
14.00	0.395	1.242	0.571	0.692	0.3870
15.00	0.397	1.257	0.562	0.698	0.3790

SPEED/POWER TRIAL PREDICTION OF SCANTLING DRAFT WITH PSV

Ship model No	:		Diameter of design propeller	:	D_{PS} =6.800m
Condition	:	Scantling draft	Pitch ratio	:	$(P/D_{PS})_{0.75R}$=0.6742
Draught (F.P)	:	T_F=14.25 m	Blade area ratio	:	A_E/A_O =0.45
Draught (A.P)	:	T_A=14.25 m	Number of blades	:	Z=4
Displacement	:	∇ =86941 m^3	Direction of rotation	:	Right
Wetted surface	:	S =11985 m^2	Number of propellers	:	1
Area of bilge keel	:	S_{BK} =110 m^2			
Proj. area above water line	:	A_T =589 m^2			

Form factor	:	1+k=1			
Roughness coefficient	:	$\Delta C_F = -0.15 \times 10^{-3}$	Appendage	:	Rudder
Air resistance coefficient	:	$C_{AA} = 0.049 \times 10^{-3}$			PSV
		Ei=1.12			

Power correction factor	:	C_P = 0.96
Revolution correction factor	:	C_N = 0.99

Table 13

V_S(kn)	t_s	w_s	η_{hs}	η_{rs}	η_{OS}	η_{DS}	P_{ES}(kW)	P_{DT}(kW)	N_T(rpm)
11.00	0.244	0.390	1.239	0.965	0.581	0.694	2407	3327	76.2
11.50	0.243	0.380	1.222	0.969	0.585	0.692	2766	3836	80.1
12.00	0.244	0.376	1.212	0.972	0.586	0.690	3147	4379	83.9
12.50	0.249	0.381	1.214	0.973	0.583	0.688	3561	4968	87.4
13.00	0.253	0.389	1.224	0.973	0.578	0.688	4027	5621	90.8
13.50	0.252	0.393	1.233	0.974	0.574	0.689	4570	6363	94.4
14.00	0.248	0.395	1.242	0.976	0.571	0.692	5210	7226	98.4
14.50	0.245	0.396	1.250	0.981	0.567	0.695	5953	8223	102.7
15.00	0.243	0.397	1.257	0.988	0.562	0.698	6802	9359	107.2

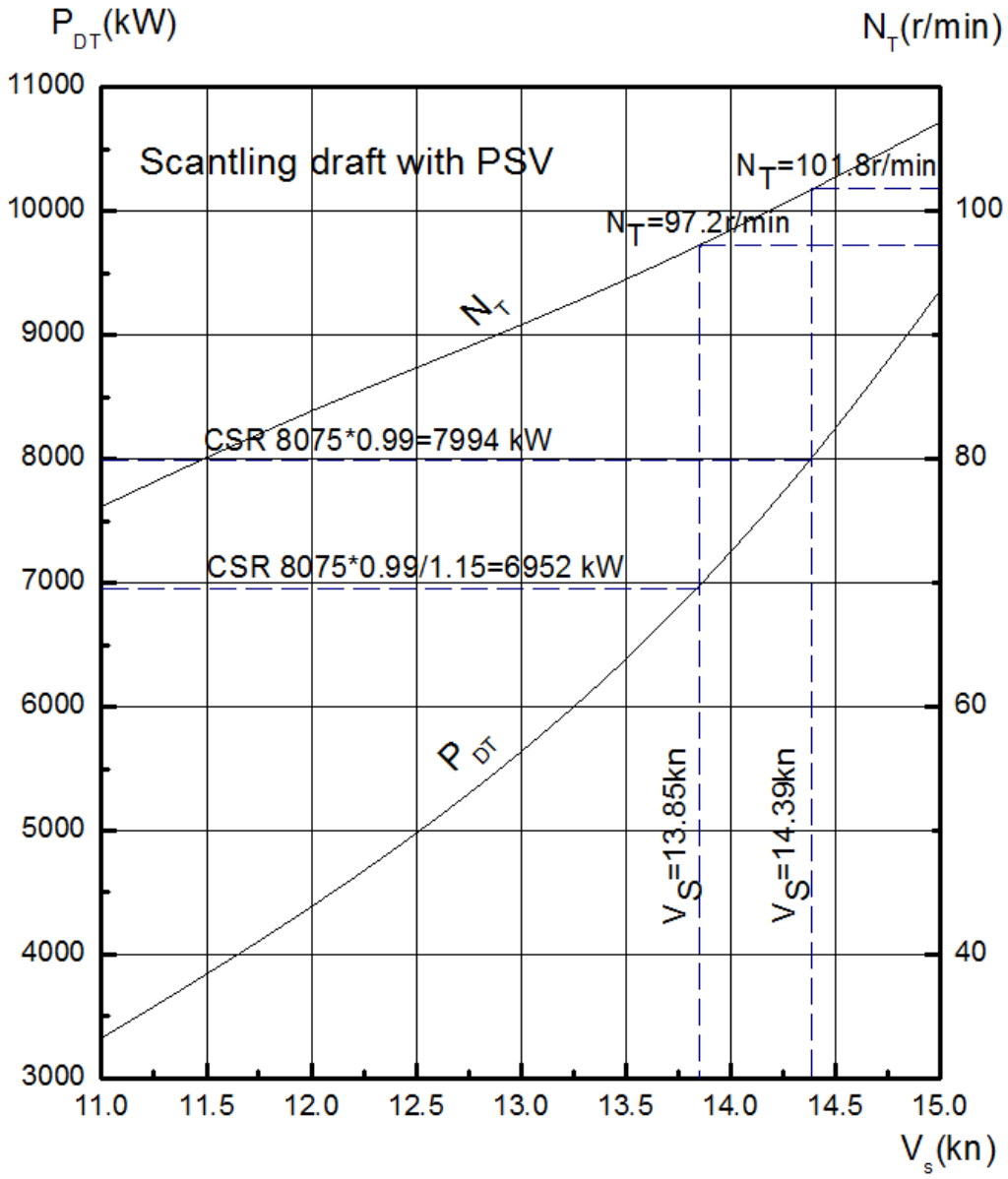


Fig. 5 SPEED/POWER TRIAL PREDICTION CURVES OF SCANTLING DRAFT WITH PSV

RESISTANCE TEST RESULTS TEST OF BALLAST DRAFT WITHOUT PSV

Ship model No	:		Length of waterline in general	:	L_{WL} =214.86 m
Condition	:	Ballast draft	Draught (F.P)	:	T_F=5.00 m
Temperature of water	:	t_m=18.6℃	Draught (A.P)	:	T_A=7.80 m
Date of tests	:	2017-5-18	Displacement	:	∇ =36134 m^3
Appendages	:	Rudder	Wetted surface	:	S =8225 m^2
			Area of bilge keel	:	S_{BK} =110 m^2
			Proj. area above water line	:	A_T =852 m^2

Form factor : 1+k=1

Roughness coefficient : $\Delta C_F = 0.05 \times 10^{-3}$

Air resistance coefficient : $C_{AA} = 0.104 \times 10^{-3}$

Table 14

V_S（kn）	F_n	C_{tm}（10^{-3}）	C_{fm}（10^{-3}）	C_{rm}（10^{-3}）
12.00	0.1345	3.9504	3.1635	0.7869
12.50	0.1401	3.9685	3.1406	0.8279
13.00	0.1457	3.9905	3.1188	0.8717
13.50	0.1513	4.0156	3.0981	0.9175
14.00	0.1569	4.0470	3.0783	0.9687
14.50	0.1625	4.0911	3.0594	1.0317
15.00	0.1681	4.1489	3.0413	1.1076
15.50	0.1737	4.2171	3.0239	1.1932
16.00	0.1793	4.2930	3.0072	1.2858

V_S（kn）	C_{rs}（10^{-3}）	C_{ts}（10^{-3}）	R_S（kN）	P_{ES}（kW）
12.00	0.7869	2.4717	397.52	2454
12.50	0.8279	2.5050	437.15	2811
13.00	0.8717	2.5414	479.70	3208
13.50	0.9175	2.5803	525.22	3647
14.00	0.9687	2.6248	574.59	4138
14.50	1.0317	2.6814	629.66	4697
15.00	1.1076	2.7512	691.36	5335
15.50	1.1932	2.8308	759.59	6056
16.00	1.2858	2.9177	834.23	6866

SELF-PROPULSION TEST RESULTS OF BALLAST DRAFT WITHOUT PSV

Ship model No	:		Length of waterline in general	:	L_{WL} =214.86 m
Propeller model No	:		Draught (F.P)	:	T_F=5.00 m
Condition	:	Ballast draft	Draught (A.P)	:	T_A=7.80 m
Temperature of water	:	t_m=18.6℃	Displacement	:	▽ =36134 m^3
Date of tests	:	2017-5-18	Wetted surface	:	S =8225 m^2
Appendages	:	Rudder	Area of bilge keel	:	S_{BK} =110 m^2
			Proj. area above water line	:	A_T =852 m^2
Form factor	:	1+k=1	Propeller diameter	:	D_{PS} =6.800m
Roughness coefficient	:	$\Delta C_F = 0.05 \times 10^{-3}$	Pitch ratio	:	$(P/D_{PS})_{0.75R}$=0.6742
Air resistance coefficient	:	$C_{AA} = 0.104 \times 10^{-3}$	Blade area ratio	:	A_E/A_O =0.45
			Number of blades	:	Z=4
			Direction of rotation	:	Right

Table 15

V_m (m/s)	R_m (N)	F_D (N)	T_m (N)	Q_m (N.m)	N_m (rps)
1.109	20.756	8.420	16.049	0.4003	6.527
1.201	24.607	9.691	19.430	0.4836	7.235
1.294	28.942	11.057	22.981	0.5724	8.013
1.386	34.061	12.483	27.620	0.6840	8.774
1.478	40.100	13.985	33.989	0.8294	9.426
V_m (m/s)	$t_m(t_s)$	w_m	$\eta_{rm}(\eta_{rs})$	η_{om}	J_t
1.109	0.231	0.512	0.962	0.550	0.3781
1.201	0.232	0.492	0.968	0.556	0.3843
1.294	0.222	0.458	0.976	0.572	0.3990
1.386	0.219	0.447	0.981	0.571	0.3980
1.478	0.232	0.480	0.979	0.542	0.3716
V_S(kn)	w_s	η_{hs}	η_{os}	η_{DS}	J_S
12.00	0.390	1.260	0.617	0.747	0.4305
13.00	0.381	1.241	0.616	0.740	0.4304
14.00	0.359	1.215	0.623	0.739	0.4372
15.00	0.353	1.207	0.620	0.735	0.4341
16.00	0.376	1.232	0.600	0.724	0.4142

SPEED/POWER TRIAL PREDICTION OF BALLAST DRAFT WITHOUT PSV

Ship model No	:		Diameter of design propeller	:	D_{PS} =6.800m
Condition	:	Ballast draft	Pitch ratio	:	$(P/D_{PS})_{0.75R}$=0.6742
Draught (F.P)	:	T_F=5.00 m	Blade area ratio	:	A_E/A_O =0.45
Draught (A.P)	:	T_A=7.80 m	Number of blades	:	Z=4
Displacement	:	∇ =36134 m^3	Direction of rotation	:	Right
Wetted surface	:	S =8225 m^2	Number of propellers	:	1
Area of bilge keel	:	S_{BK} =110 m^2			
Proj. area above water line	:	A_T =852 m^2			

Form factor	:	1+k=1			
Roughness coefficient	:	$\Delta C_F = 0.05 \times 10^{-3}$	Appendage	:	Rudder
Air resistance coefficient	:	$C_{AA} = 0.104 \times 10^{-3}$			

Power correction factor	:	C_P = 1.0
Revolution correction factor	:	C_N = 1.0

Table 16

V_S(kn)	t_s	w_s	η_{hs}	η_{rs}	η_{OS}	η_{DS}	P_{ES}(kW)	P_{DT}(kW)	N_T(rpm)
12.00	0.231	0.390	1.260	0.962	0.617	0.747	2454	3285	77.2
12.50	0.233	0.387	1.251	0.964	0.616	0.743	2811	3784	81.0
13.00	0.232	0.381	1.241	0.968	0.616	0.740	3208	4335	84.8
13.50	0.228	0.371	1.228	0.972	0.620	0.739	3647	4934	88.9
14.00	0.222	0.359	1.215	0.976	0.623	0.739	4138	5599	93.1
14.50	0.218	0.353	1.207	0.980	0.624	0.738	4697	6365	97.3
15.00	0.219	0.353	1.207	0.981	0.620	0.735	5335	7260	101.5
15.50	0.224	0.362	1.217	0.981	0.612	0.730	6056	8299	105.5
16.00	0.232	0.376	1.232	0.979	0.600	0.724	6866	9488	109.4

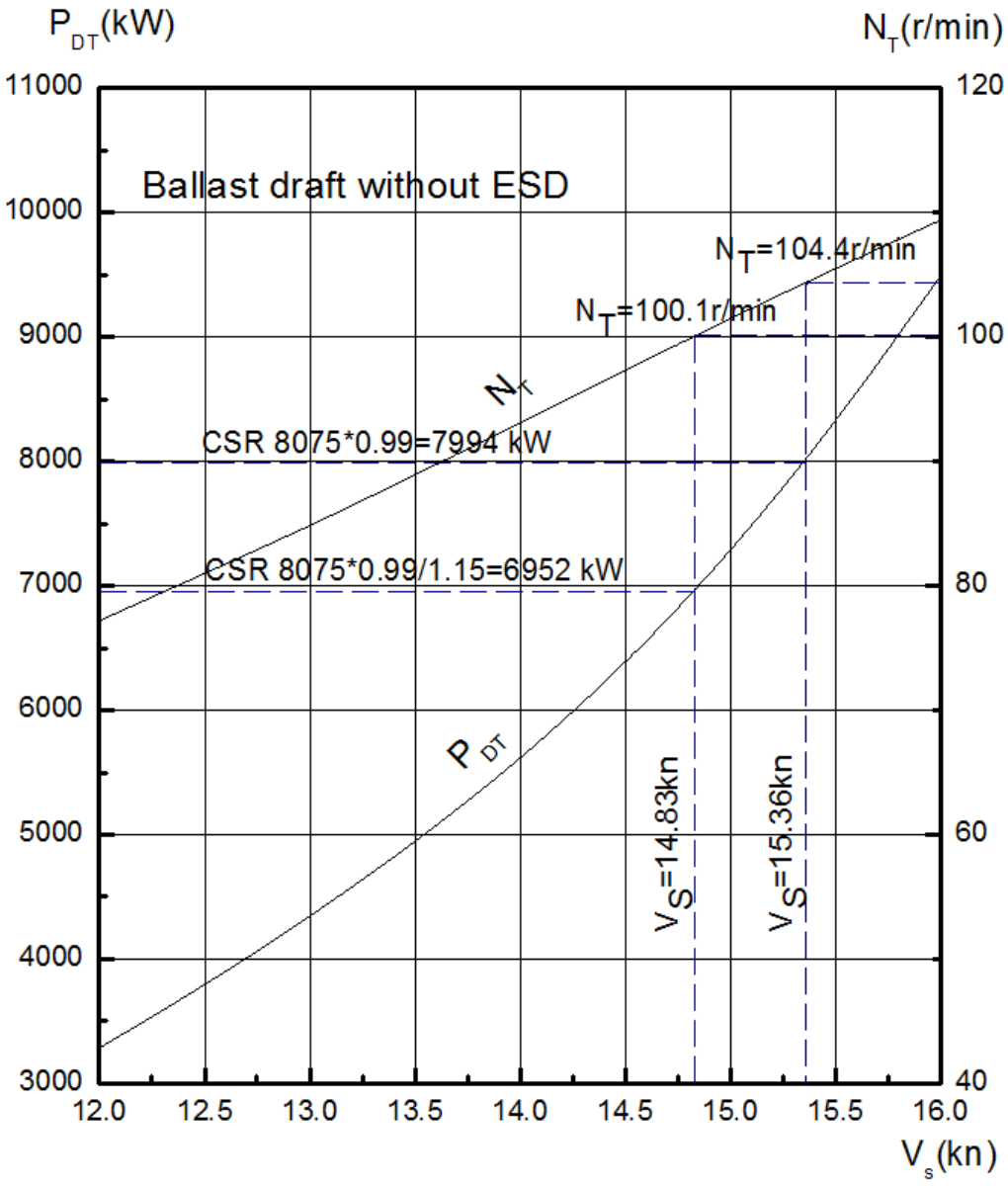


Fig. 6 SPEED/POWER TRIAL PREDICTION CURVES OF BALLAST DRAFT WITHOUT PSV

RESISTANCE TEST RESULTS TEST OF BALLAST DRAFT WITH PSV

Ship model No	:		Length of waterline in general	:	L_{WL} =214.86 m
Condition	:	Ballast draft	Draught (F.P)	:	T_F=5.00 m
Temperature of water	:	t_m=18.6℃	Draught (A.P)	:	T_A=7.80 m
Date of tests	:	2017-5-18	Displacement	:	▽ =36134 m^3
Appendages	:	Rudder	Wetted surface	:	S =8282 m^2
		PSV	Area of bilge keel	:	S_{BK} =110 m^2
			Proj. area above water line	:	A_T =852 m^2

Form factor	:	1+k=1
Roughness coefficient	:	$\Delta C_F = 0.05 \times 10^{-3}$
Air resistance coefficient	:	$C_{AA} = 0.103 \times 10^{-3}$

Table 17

V_S（kn）	F_n	C_{tm}（10^{-3}）	C_{fm}（10^{-3}）	C_{rm}（10^{-3}）
12.00	0.1345	3.8849	3.1635	0.7214
12.50	0.1401	3.9231	3.1406	0.7825
13.00	0.1457	3.9482	3.1188	0.8294
13.50	0.1513	3.9687	3.0981	0.8706
14.00	0.1569	3.9927	3.0783	0.9144
14.50	0.1625	4.0277	3.0594	0.9683
15.00	0.1681	4.0779	3.0413	1.0366
15.50	0.1737	4.1446	3.0239	1.1207
16.00	0.1793	4.2270	3.0072	1.2198

V_S（kn）	C_{rs}（10^{-3}）	C_{ts}（10^{-3}）	R_S（kN）	P_{ES}（kW）
12.00	0.7214	2.4053	389.53	2404
12.50	0.7825	2.4587	432.05	2778
13.00	0.8294	2.4983	474.83	3175
13.50	0.8706	2.5325	519.08	3605
14.00	0.9144	2.5697	566.42	4079
14.50	0.9683	2.6172	618.83	4616
15.00	1.0366	2.6793	677.97	5231
15.50	1.1207	2.7575	745.04	5940
16.00	1.2198	2.8508	820.76	6755

SELF-PROPULSION TEST RESULTS OF BALLAST DRAFT WITH PSV

Ship model No	:		Length of waterline in general	:	L_{WL} =214.86 m
Propeller model No	:		Draught (F.P)	:	T_F=5.00 m
Condition	:	Ballast draft	Draught (A.P)	:	T_A=7.80 m
Temperature of water	:	t_m=18.6℃	Displacement	:	▽ =36134 m^3
Date of tests	:	2017-5-18	Wetted surface	:	S =8282 m^2
Appendages	:	Rudder	Area of bilge keel	:	S_{BK} =110 m^2
		PSV	Proj. area above water line	:	A_T =852 m^2
Form factor	:	1+k=1	Propeller diameter	:	D_{PS} =6.800m
Roughness coefficient	:	$\Delta C_F = 0.05 \times 10^{-3}$	Pitch ratio	:	$(P/D_{PS})_{0.75R}$=0.6742
Air resistance coefficient	:	$C_{AA} = 0.103 \times 10^{-3}$	Blade area ratio	:	A_E/A_O =0.45
		Ei=1.171	Number of blades	:	Z=4
			Direction of rotation	:	Right

Table 18

V_m (m/s)	R_m (N)	F_D (N)	T_m (N)	Q_m (N.m)	N_m (rps)
1.109	20.553	8.479	15.930	0.3951	6.395
1.201	24.515	9.759	19.539	0.4819	6.974
1.294	28.752	11.134	23.550	0.5777	7.671
1.386	33.710	12.570	28.141	0.6800	8.462
1.478	39.757	14.082	33.791	0.8129	9.323
V_m (m/s)	$t_m(t_s)$	w_m	$\eta_{rm}(\eta_{rs})$	η_{om}	J_t
1.109	0.242	0.540	0.958	0.534	0.3638
1.201	0.245	0.554	0.956	0.518	0.3501
1.294	0.252	0.543	0.962	0.520	0.3518
1.386	0.249	0.518	0.981	0.529	0.3599
1.478	0.240	0.495	0.989	0.535	0.3647
V_S(kn)	w_s	η_{hs}	η_{os}	η_{DS}	J_S
12.00	0.461	1.406	0.583	0.786	0.3982
13.00	0.478	1.446	0.568	0.786	0.3846
14.00	0.464	1.396	0.570	0.766	0.3864
15.00	0.436	1.331	0.580	0.757	0.3951
16.00	0.409	1.286	0.586	0.744	0.4004

SPEED/POWER TRIAL PREDICTION OF BALLAST DRAFT WITH PSV

Ship model No	:		Diameter of design propeller	:	D_{PS} =6.800m
Condition	:	Ballast draft	Pitch ratio	:	$(P/D_{PS})_{0.75R}$=0.6742
Draught (F.P)	:	T_F=5.00 m	Blade area ratio	:	A_E/A_O =0.45
Draught (A.P)	:	T_A=7.80 m	Number of blades	:	Z=4
Displacement	:	∇ =36134 m^3	Direction of rotation	:	Right
Wetted surface	:	S =8282 m^2	Number of propellers	:	1
Area of bilge keel	:	S_{BK} =110 m^2			
Proj. area above water line	:	A_T =852 m^2			

Form factor	:	1+k=1			
Roughness coefficient	:	$\Delta C_F = 0.05 \times 10^{-3}$	Appendage	:	Rudder
Air resistance coefficient	:	$C_{AA} = 0.104 \times 10^{-3}$			PSV
		Ei=1.171			

Power correction factor	:	C_P = 1.0
Revolution correction factor	:	C_N = 1.0

Table 19

V_S(kn)	t_s	w_s	η_{hs}	η_{rs}	η_{OS}	η_{DS}	P_{ES}(kW)	P_{DT}(kW)	N_T(rpm)
12.00	0.242	0.461	1.406	0.958	0.583	0.786	2404	3059	73.7
12.50	0.243	0.472	1.435	0.957	0.574	0.788	2778	3525	76.8
13.00	0.245	0.478	1.446	0.956	0.568	0.786	3175	4042	80.1
13.50	0.249	0.475	1.430	0.957	0.567	0.776	3605	4643	83.9
14.00	0.252	0.464	1.396	0.962	0.570	0.766	4079	5324	88.1
14.50	0.252	0.450	1.362	0.972	0.575	0.761	4616	6067	92.6
15.00	0.249	0.436	1.331	0.981	0.580	0.757	5231	6907	97.3
15.50	0.245	0.422	1.307	0.987	0.583	0.752	5940	7902	102.2
16.00	0.240	0.409	1.286	0.989	0.586	0.744	6755	9074	107.2

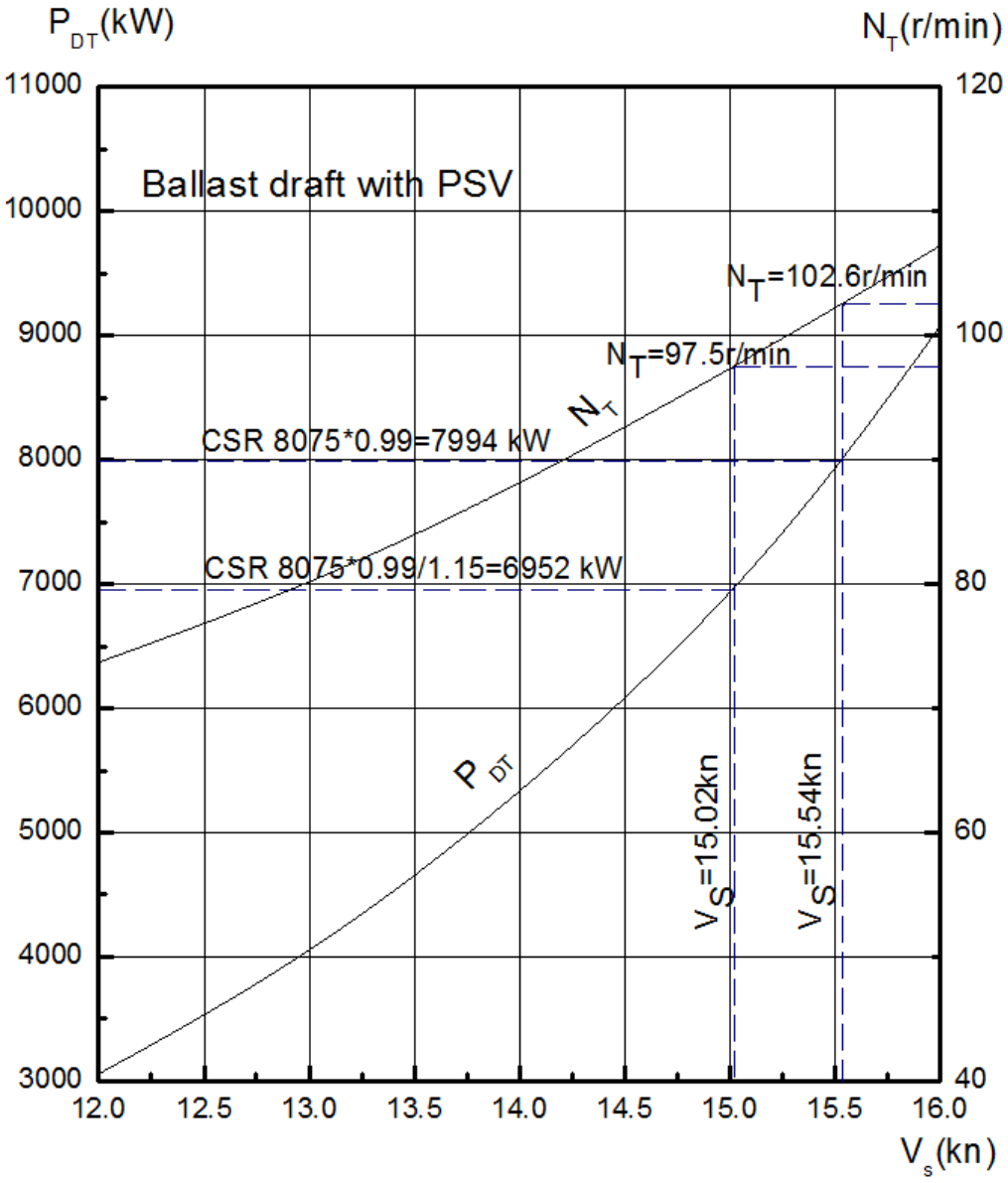


Fig. 7 SPEED/POWER TRIAL PREDICTION CURVES OF BALLAST DRAFT WITH PSV

RESISTANCE TEST RESULTS TEST OF HEAVY BALLAST DRAFT WITHOUT PSV

Ship model No	:		Length of waterline in general	:	L_{WL} =217.91 m
Condition	:	Heavy ballast draft	Draught (F.P)	:	T_F=4.40 m
Temperature of water	:	t_m=18.6℃	Draught (A.P)	:	T_A=9.40 m
Date of tests	:	2017-5-18	Displacement	:	▽ =39024 m^3
Appendages	:	Rudder	Wetted surface	:	S =8485 m^2
			Area of bilge keel	:	S_{BK} =110 m^2
			Proj. area above water line	:	A_T =836 m^2

Form factor	:	1+k=1
Roughness coefficient	:	$\Delta C_F = 0.05 \times 10^{-3}$
Air resistance coefficient	:	$C_{AA} = 0.099 \times 10^{-3}$

Table 20

V_S（kn）	F_n	C_{tm}（10^{-3}）	C_{fm}（10^{-3}）	C_{rm}（10^{-3}）
12.00	0.1335	3.9251	3.1556	0.7695
12.50	0.1391	3.9533	3.1328	0.8205
13.00	0.1447	3.9800	3.1111	0.8689
13.50	0.1502	4.0094	3.0904	0.9190
14.00	0.1558	4.0432	3.0707	0.9725
14.50	0.1614	4.0816	3.0518	1.0298
15.00	0.1669	4.1269	3.0338	1.0931
15.50	0.1725	4.1832	3.0165	1.1667
16.00	0.1780	4.2516	2.9999	1.2517

V_S（kn）	C_{rs}（10^{-3}）	C_{ts}（10^{-3}）	R_S（kN）	P_{ES}（kW）
12.00	0.7695	2.4459	405.82	2505
12.50	0.8205	2.4893	448.15	2882
13.00	0.8689	2.5304	492.72	3295
13.50	0.9190	2.5735	540.40	3753
14.00	0.9725	2.6204	591.75	4262
14.50	1.0298	2.6712	647.09	4827
15.00	1.0931	2.7284	707.32	5458
15.50	1.1667	2.7961	774.00	6171
16.00	1.2517	2.8755	848.14	6981

SELF-PROPULSION TEST RESULTS OF HEAVY BALLAST DRAFT WITHOUT PSV

Ship model No	:		Length of waterline in general	:	L_{WL} =217.91 m
Propeller model No	:		Draught (F.P)	:	T_F=4.40 m
Condition	:	Heavy ballast draft	Draught (A.P)	:	T_A=9.40 m
Temperature of water	:	t_m=18.6℃	Displacement	:	▽ =39024 m^3
Date of tests	:	2017-5-18	Wetted surface	:	S =8485 m^2
Appendages	:	Rudder	Area of bilge keel	:	S_{BK} =110 m^2
			Proj. area above water line	:	A_T =836 m^2

Form factor	:	1+k=1	Propeller diameter	:	D_{PS} =6.800m
Roughness coefficient	:	$\Delta C_F = 0.05 \times 10^{-3}$	Pitch ratio	:	$(P/D_{PS})_{0.75R}$=0.6742
Air resistance coefficient	:	$C_{AA} = 0.099 \times 10^{-3}$	Blade area ratio	:	A_E/A_O =0.45
			Number of blades	:	Z=4
			Direction of rotation	:	Right

Table 21

V_m (m/s)	R_m (N)	F_D (N)	T_m (N)	Q_m (N.m)	N_m (rps)
1.109	21.275	8.658	16.171	0.4103	6.737
1.201	25.318	9.965	19.769	0.4983	7.455
1.294	29.829	11.369	23.830	0.5959	8.150
1.386	34.952	12.836	28.529	0.7079	8.854
1.478	40.968	14.380	34.131	0.8397	9.592
V_m (m/s)	$t_m(t_s)$	w_m	$\eta_{rm}(\eta_{rs})$	η_{om}	J_t
1.109	0.220	0.466	0.960	0.574	0.4007
1.201	0.223	0.453	0.966	0.574	0.4014
1.294	0.225	0.450	0.972	0.571	0.3980
1.386	0.225	0.450	0.976	0.565	0.3923
1.478	0.221	0.453	0.979	0.557	0.3845
V_S(kn)	w_s	η_{hs}	η_{os}	η_{DS}	J_S
12.00	0.362	1.222	0.627	0.736	0.4419
13.00	0.358	1.209	0.624	0.729	0.4384
14.00	0.357	1.205	0.619	0.726	0.4335
15.00	0.358	1.207	0.614	0.723	0.4281
16.00	0.357	1.212	0.608	0.721	0.4219

SPEED/POWER TRIAL PREDICTION OF HEAVY BALLAST DRAFT WITHOUT PSV

Ship model No	:		Diameter of design propeller	:	D_{PS} =6.800m
Condition	:	Heavy ballast draft	Pitch ratio	:	$(P/D_{PS})_{0.75R}$=0.6742
Draught (F.P)	:	T_F=4.40 m	Blade area ratio	:	A_E/A_O =0.45
Draught (A.P)	:	T_A=9.40 m	Number of blades	:	Z=4
Displacement	:	∇ =39024 m^3	Direction of rotation	:	Right
Wetted surface	:	S =8485 m^2	Number of propellers	:	1
Area of bilge keel	:	S_{BK} =110 m^2			
Proj. area above water line	:	A_T =836 m^2			

Form factor	:	1+k=1			
Roughness coefficient	:	$\Delta C_F = 0.05 \times 10^{-3}$	Appendage	:	Rudder
Air resistance coefficient	:	$C_{AA} = 0.099 \times 10^{-3}$			

Power correction factor	:	C_P = 1.0
Revolution correction factor	:	C_N = 1.0

Table 22

V_S(kn)	t_s	w_s	η_{hs}	η_{rs}	η_{OS}	η_{DS}	P_{ES}(kW)	P_{DT}(kW)	N_T(rpm)
12.00	0.220	0.362	1.222	0.960	0.627	0.736	2505	3404	78.7
12.50	0.222	0.359	1.215	0.963	0.626	0.732	2882	3935	82.6
13.00	0.223	0.358	1.209	0.966	0.624	0.729	3295	4518	86.5
13.50	0.225	0.357	1.206	0.969	0.622	0.727	3753	5160	90.3
14.00	0.225	0.357	1.205	0.972	0.619	0.726	4262	5872	94.2
14.50	0.225	0.358	1.206	0.974	0.617	0.724	4827	6663	98.1
15.00	0.225	0.358	1.207	0.976	0.614	0.723	5458	7547	102.2
15.50	0.223	0.357	1.209	0.977	0.611	0.722	6171	8545	106.4
16.00	0.221	0.357	1.212	0.979	0.608	0.721	6981	9677	110.7

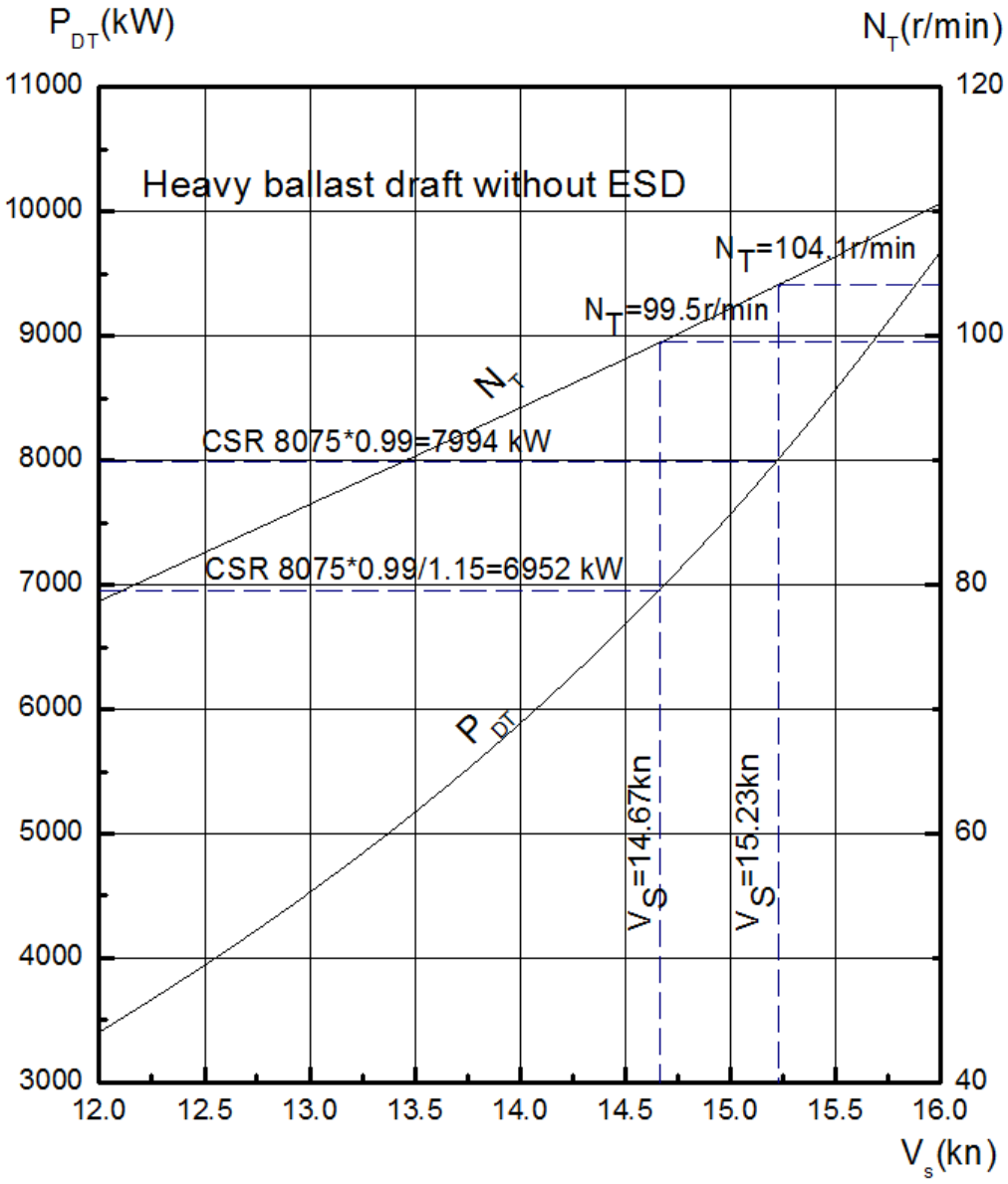


Fig. 8 SPEED/POWER TRIAL PREDICTION CURVES OF HEAVY BALLAST DRAFT WITHOUT PSV

RESISTANCE TEST RESULTS TEST OF HEAVY BALLAST DRAFT WITH PSV

Ship model No	:		Length of waterline in general	:	L_{WL} =217.91 m
Condition	:	Heavy ballast draft	Draught (F.P)	:	T_F=4.40 m
Temperature of water	:	t_m=20.6℃	Draught (A.P)	:	T_A=9.40 m
Date of tests	:	2017-6-7	Displacement	:	▽ =39024 m^3
Appendages	:	Rudder	Wetted surface	:	S =8542 m^2
		PSV	Area of bilge keel	:	S_{BK} =110 m^2
			Proj. area above water line	:	A_T =836 m^2

Form factor	:	1+k=1
Roughness coefficient	:	$\Delta C_F = 0.05 \times 10^{-3}$
Air resistance coefficient	:	$C_{AA} = 0.098 \times 10^{-3}$

Table 23

V_S（kn）	F_n	C_{tm}（10^{-3}）	C_{fm}（10^{-3}）	C_{rm}（10^{-3}）
12.00	0.1335	3.8618	3.1283	0.7335
12.50	0.1391	3.9204	3.1057	0.8147
13.00	0.1447	3.9568	3.0843	0.8725
13.50	0.1502	3.9728	3.0639	0.9089
14.00	0.1558	3.9792	3.0445	0.9347
14.50	0.1614	3.9932	3.0258	0.9674
15.00	0.1669	4.0256	3.0080	1.0176
15.50	0.1725	4.0821	2.9909	1.0912
16.00	0.1780	4.1631	2.9745	1.1886

V_S（kn）	C_{rs}（10^{-3}）	C_{ts}（10^{-3}）	R_S（kN）	P_{ES}（kW）
12.00	0.7335	2.4092	402.40	2484
12.50	0.8147	2.4826	449.95	2893
13.00	0.8725	2.5332	496.57	3321
13.50	0.9089	2.5626	541.72	3762
14.00	0.9347	2.5818	586.96	4227
14.50	0.9674	2.6080	636.03	4744
15.00	1.0176	2.6521	692.15	5341
15.50	1.0912	2.7198	757.92	6043
16.00	1.1886	2.8115	834.85	6871

SELF-PROPULSION TEST RESULTS OF HEAVY BALLAST DRAFT WITHOUT PSV

Ship model No	:		Length of waterline in general	:	L_{WL} =217.91 m
Propeller model No	:		Draught (F.P)	:	T_F=4.40 m
Condition	:	Heavy ballast draft	Draught (A.P)	:	T_A=9.40 m
Temperature of water	:	t_m=20.6℃	Displacement	:	▽ =39024 m^3
Date of tests	:	2017-6-7	Wetted surface	:	S =8542 m^2
Appendages	:	Rudder	Area of bilge keel	:	S_{BK} =110 m^2
		PSV	Proj. area above water line	:	A_T =836 m^2
Form factor	:	1+k=1	Propeller diameter	:	D_{PS} =6.800m
Roughness coefficient	:	$\Delta C_F = 0.05 \times 10^{-3}$	Pitch ratio	:	$(P/D_{PS})_{0.75R}$=0.6742
Air resistance coefficient	:	$C_{AA} = 0.098 \times 10^{-3}$	Blade area ratio	:	A_E/A_O =0.45
		Ei=1.181	Number of blades	:	Z=4
			Direction of rotation	:	Right

Table 24

V_m (m/s)	R_m (N)	F_D (N)	T_m (N)	Q_m (N.m)	N_m (rps)
1.109	21.064	8.564	16.101	0.4034	6.607
1.201	25.329	9.857	19.470	0.4828	7.238
1.294	29.543	11.246	23.421	0.5757	7.906
1.386	34.309	12.697	28.159	0.6871	8.623
1.478	40.369	14.225	33.879	0.8221	9.402
V_m (m/s)	$t_m(t_s)$	w_m	$\eta_{rm}(\eta_{rs})$	η_{om}	J_t
1.109	0.224	0.495	0.963	0.559	0.3867
1.201	0.205	0.493	0.971	0.556	0.3836
1.294	0.219	0.490	0.977	0.552	0.3802
1.386	0.233	0.487	0.981	0.547	0.3758
1.478	0.228	0.483	0.984	0.541	0.3707
V_S(kn)	w_s	η_{hs}	η_{os}	η_{DS}	J_S
12.00	0.403	1.301	0.611	0.765	0.4244
13.00	0.401	1.327	0.608	0.783	0.4217
14.00	0.398	1.298	0.604	0.766	0.4181
15.00	0.394	1.267	0.600	0.746	0.4139
16.00	0.389	1.263	0.595	0.739	0.4091

SPEED/POWER TRIAL PREDICTION OF HEAVY BALLAST DRAFT WITHOUT PSV

Ship model No	:		Diameter of design propeller	:	D_{PS} =6.800m
Condition	:	Heavy ballast draft	Pitch ratio	:	$(P/D_{PS})_{0.75R}$=0.6742
Draught (F.P)	:	T_F=4.40 m	Blade area ratio	:	A_E/A_O =0.45
Draught (A.P)	:	T_A=9.40 m	Number of blades	:	Z=4
Displacement	:	∇ =39024 m^3	Direction of rotation	:	Right
Wetted surface	:	S =8542 m^2	Number of propellers	:	1
Area of bilge keel	:	S_{BK} =110 m^2			
Proj. area above water line	:	A_T =836 m^2			

Form factor	:	1+k=1			
Roughness coefficient	:	$\Delta C_F = 0.05 \times 10^{-3}$	Appendage	:	Rudder
Air resistance coefficient	:	$C_{AA} = 0.098 \times 10^{-3}$			PSV
		Ei=1.181			

Power correction factor	:	C_P = 1.0
Revolution correction factor	:	C_N = 1.0

Table 25

V_S(kn)	t_s	w_s	η_{hs}	η_{rs}	η_{OS}	η_{DS}	P_{ES}(kW)	P_{DT}(kW)	N_T(rpm)
12.00	0.224	0.403	1.301	0.963	0.611	0.765	2484	3249	76.6
12.50	0.211	0.402	1.319	0.967	0.609	0.777	2893	3723	80.2
13.00	0.205	0.401	1.327	0.971	0.608	0.783	3321	4242	83.8
13.50	0.209	0.400	1.317	0.974	0.606	0.778	3762	4837	87.6
14.00	0.219	0.398	1.298	0.977	0.604	0.766	4227	5515	91.5
14.50	0.228	0.396	1.280	0.980	0.602	0.755	4744	6284	95.5
15.00	0.233	0.394	1.267	0.981	0.600	0.746	5341	7158	99.6
15.50	0.232	0.392	1.263	0.983	0.597	0.741	6043	8150	104.0
16.00	0.228	0.389	1.263	0.984	0.595	0.739	6871	9294	108.4

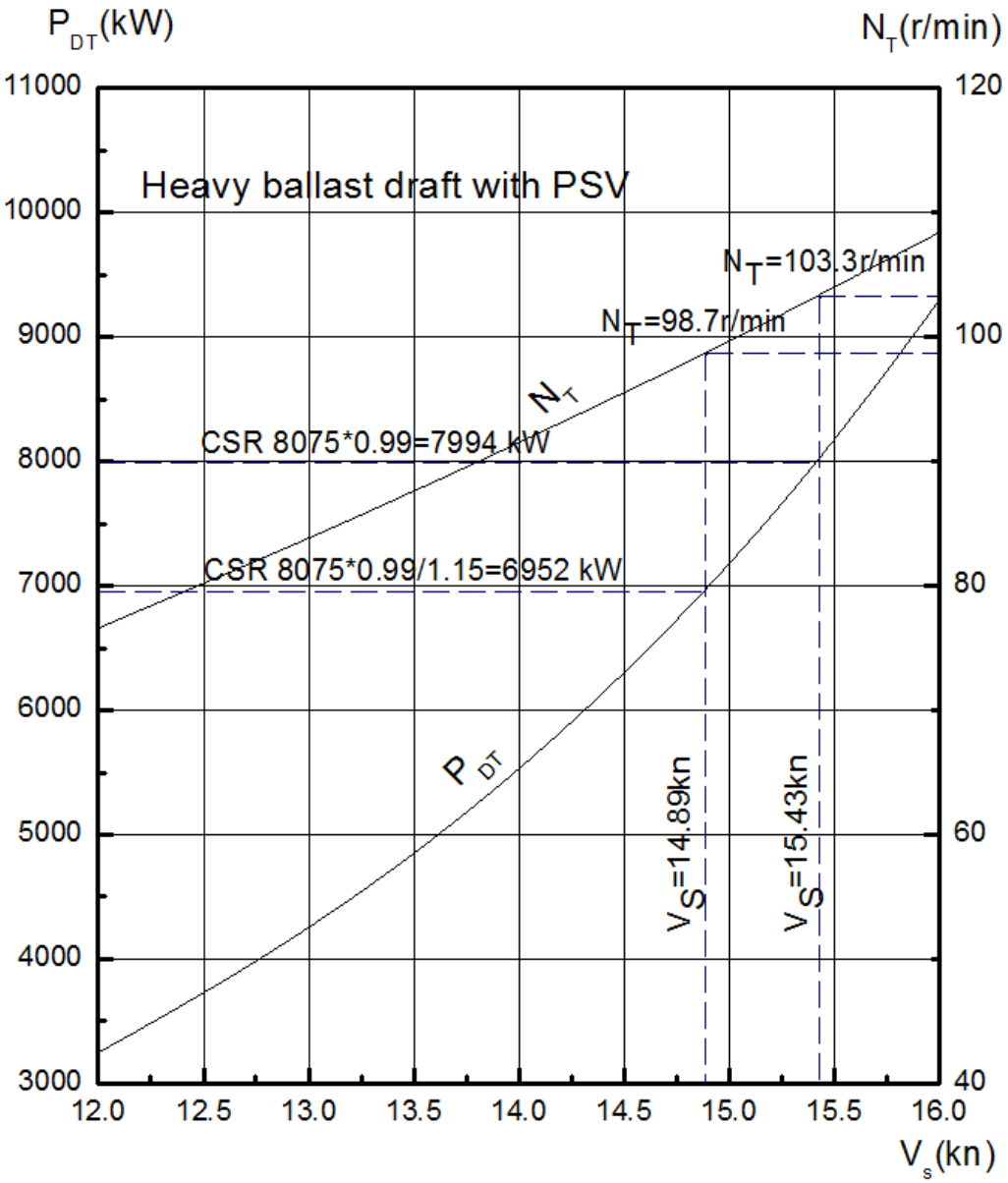
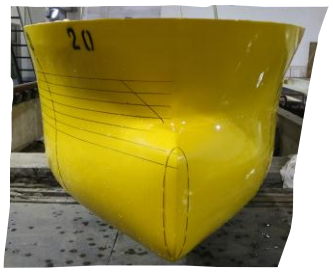


Fig. 9 SPEED/POWER TRIAL PREDICTION CURVES OF HEAVY BALLAST DRAFT WITH PSV

PHOTOGRAPHS

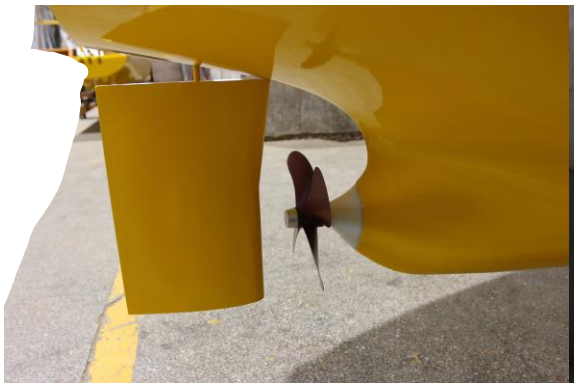


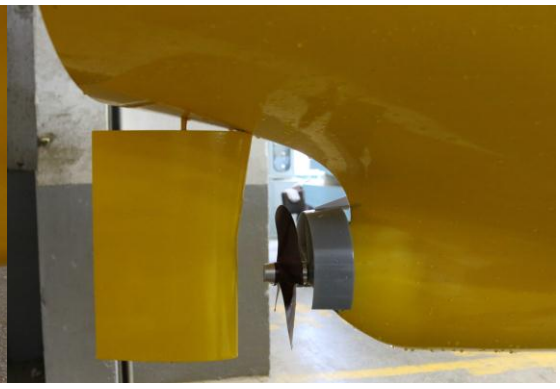




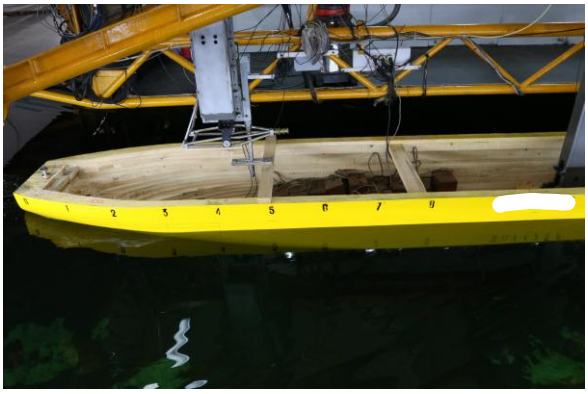
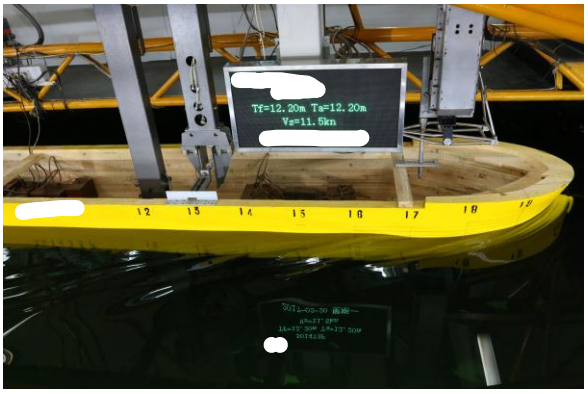


SHIP

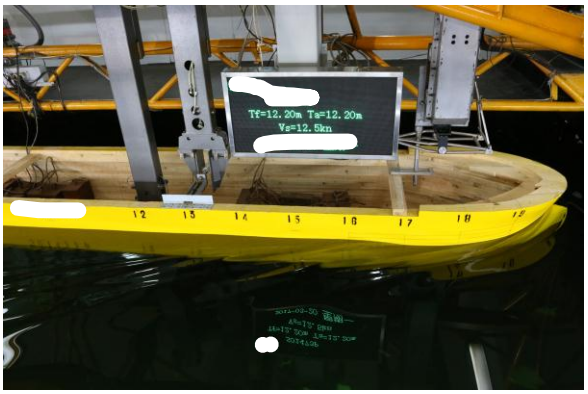




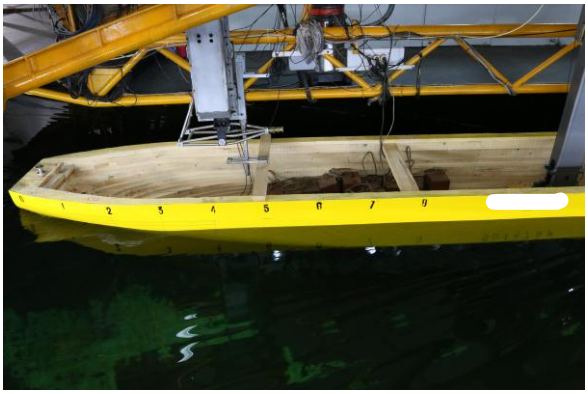
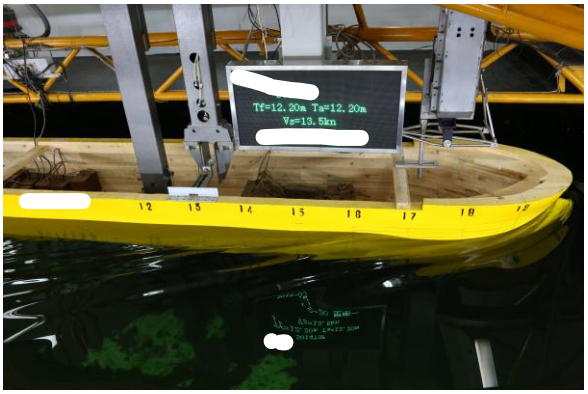
PROPELLER AND PSV

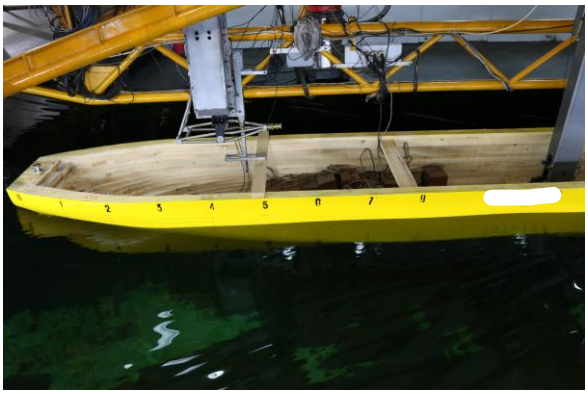
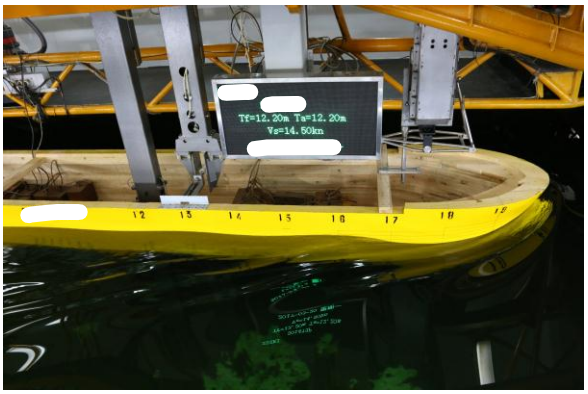
Design draft T_F=12.20m/T_A=12.20m V_S=11.50 kn

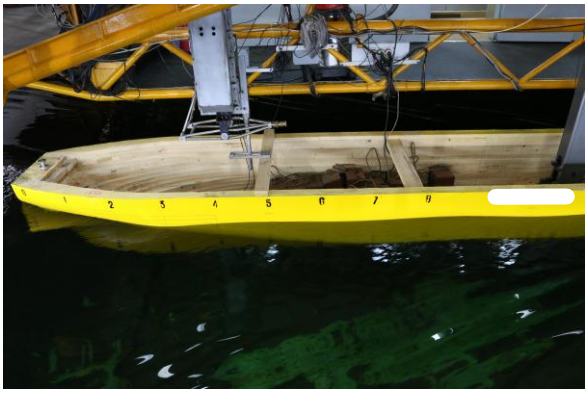
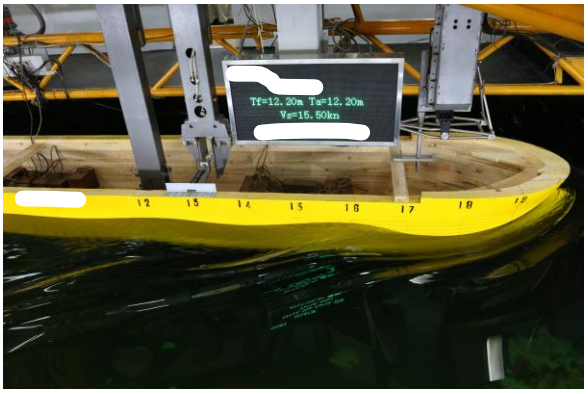
Design draft T_F=12.20m/T_A=12.20m V_S=12.50 kn

Design draft T_F=12.20m/T_A=12.20m V_S=13.50 kn

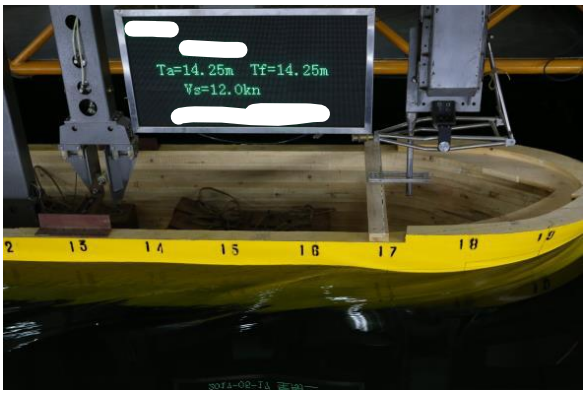
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Design draft T_F=12.20m/T_A=12.20m V_S=15.50 kn



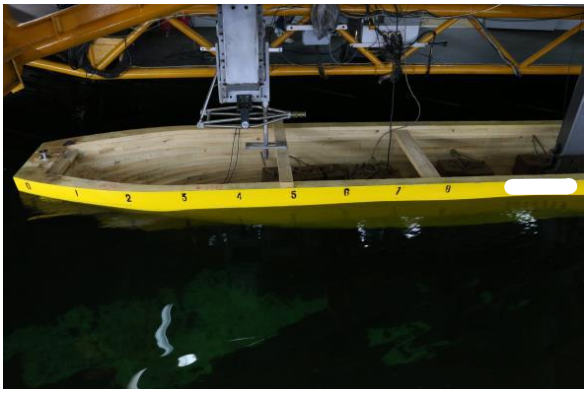

Scantling draft T_F=14.25m/T_A=14.25m V_S=11.00 kn

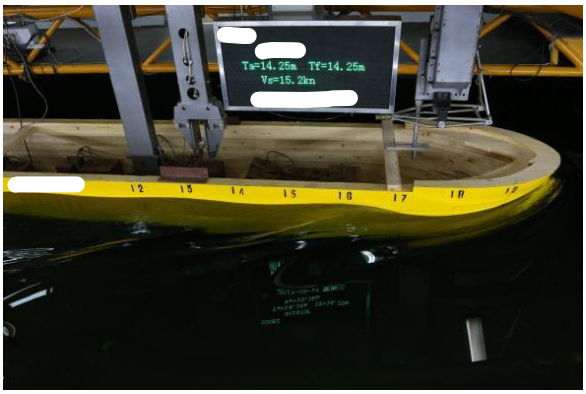
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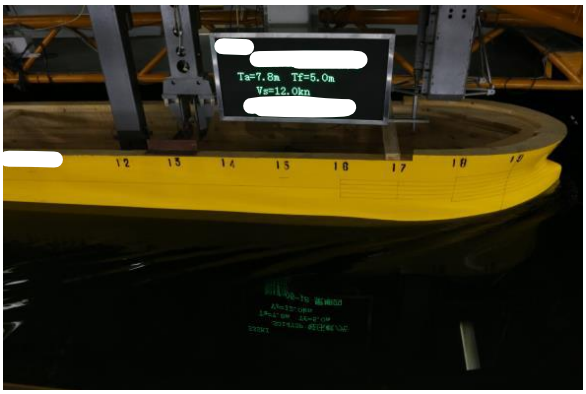

Scantling draft T_F=14.25m/T_A=14.25m V_S=13.20 kn



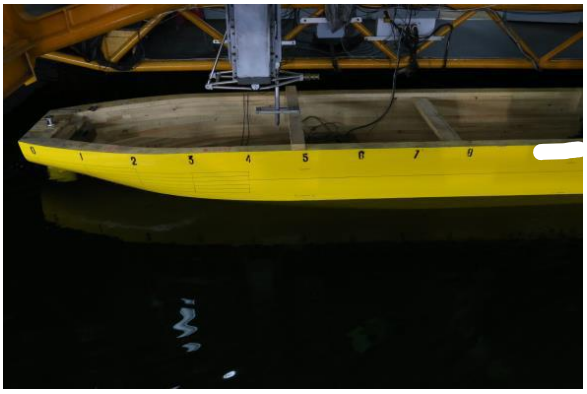
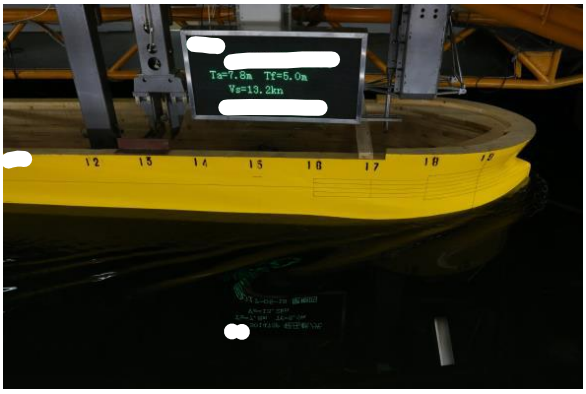

Scantling draft T_F=14.25m/T_A=14.25m V_S=14.20 kn

Scantling draft T_F=14.25m/T_A=14.25m V_S=15.20 kn

Ballast draft T_F=5.00m/T_A=7.80m V_S=12.00 kn

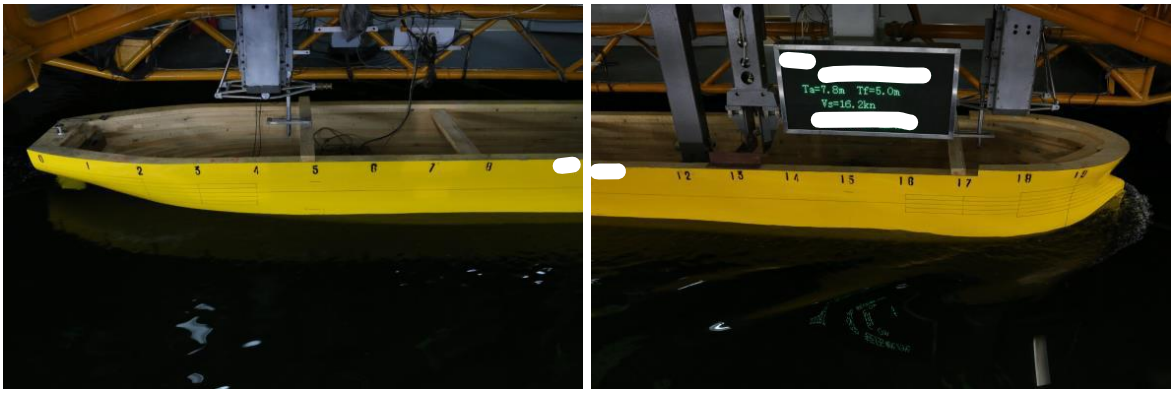
Ballast draft T_F=5.00m/T_A=7.80m V_S=13.20 kn



Ballast draft T_F=5.00m/T_A=7.80m V_S=14.20 kn



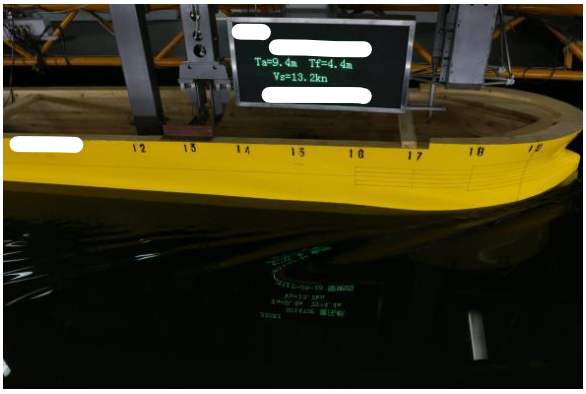
Ballast draft T_F=5.00m/T_A=7.80m V_S=15.20 kn



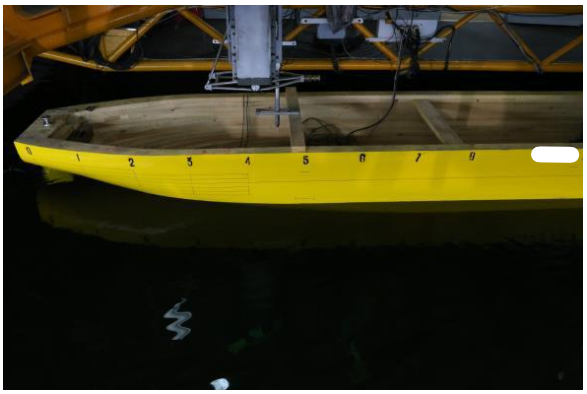
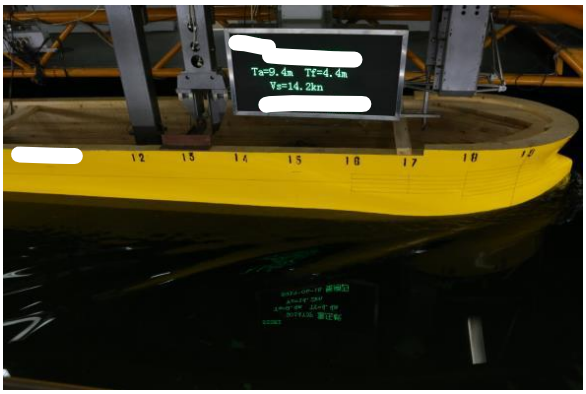
Ballast draft T_F=5.00m/T_A=7.80m V_S=16.20 kn



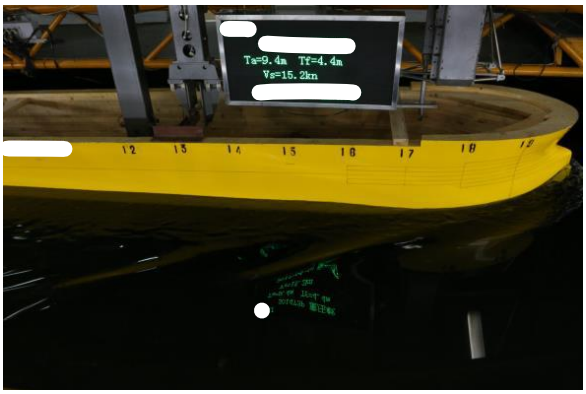
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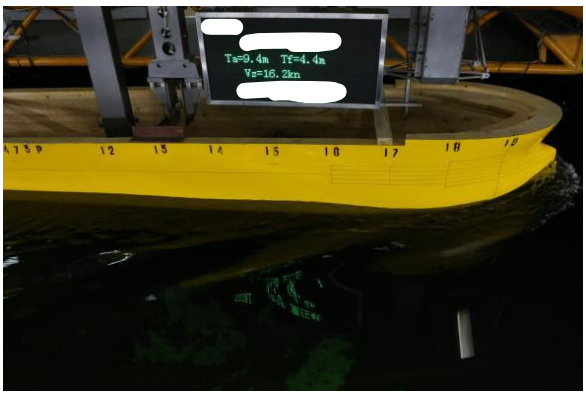
Heavy ballast draft T_F=4.40m/T_A=9.40m V_S=13.20 kn

Heavy ballast draft T_F=4.40m/T_A=9.40m V_S=14.20 kn

Heavy ballast draft T_F=4.40m/T_A=9.40m V_S=15.20 kn

Heavy ballast draft T_F=4.40m/T_A=9.40m V_S=16.20 kn

PHOTOGRAPHS OF WAVE PATTERN